

# git log/history

## References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

## Git History - Directory Commit Statistics

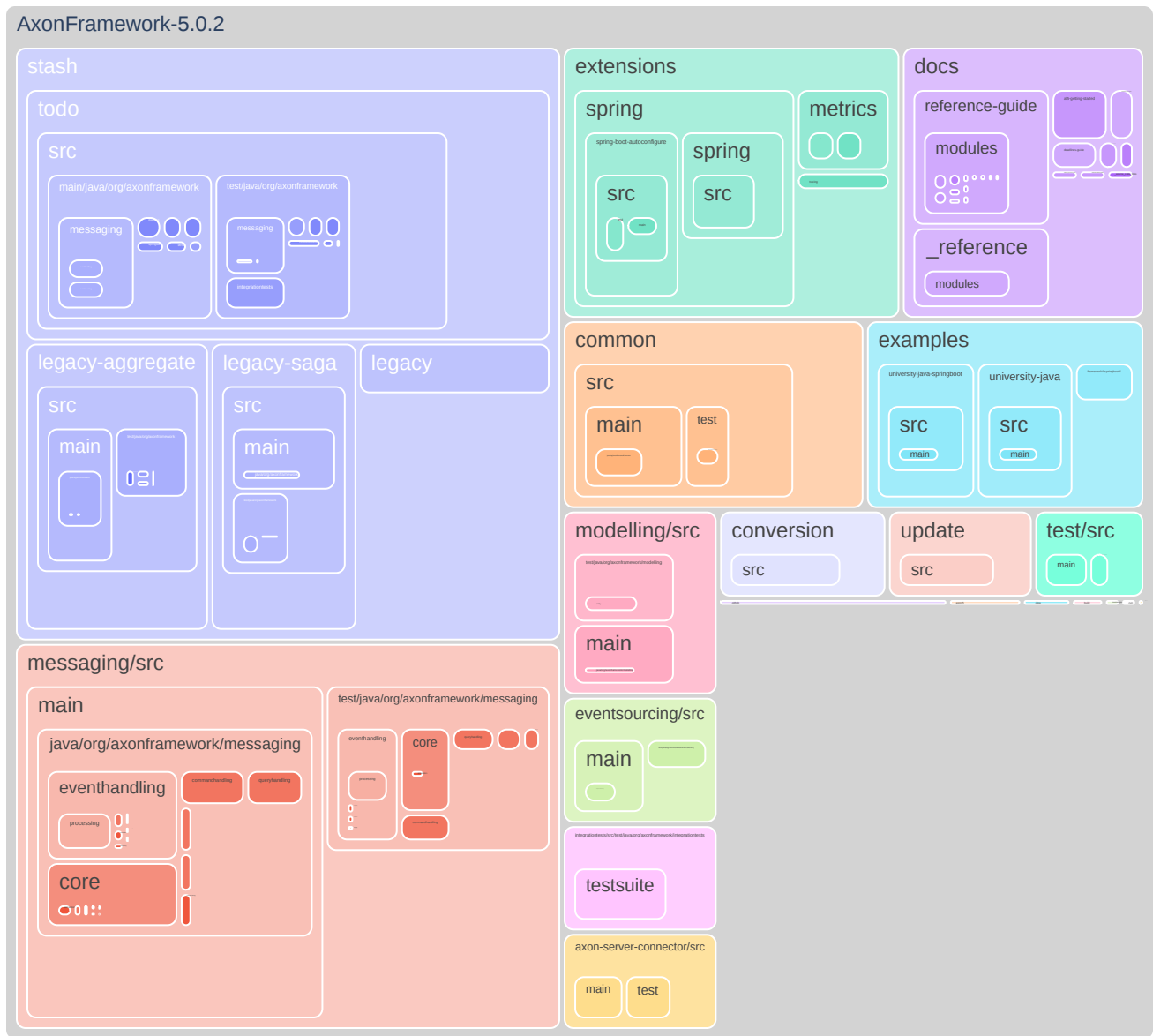
### Data Preview

|       | fileCount   | authorCount | commitCount | daysSinceLastCommit | daysSinceLastCreation | daysSinceLastModification |
|-------|-------------|-------------|-------------|---------------------|-----------------------|---------------------------|
| count | 725.000000  | 725.000000  | 725.000000  | 725.000000          | 725.000000            | 725.000000                |
| mean  | 22.416552   | 11.197241   | 171.793103  | 61.124138           | 131.329655            | 101.489655                |
| std   | 112.370074  | 12.384873   | 363.609595  | 146.368022          | 185.138332            | 152.138727                |
| min   | 1.000000    | 2.000000    | 3.000000    | 9.000000            | 11.000000             | 9.000000                  |
| 25%   | 2.000000    | 4.000000    | 22.000000   | 15.000000           | 71.000000             | 18.000000                 |
| 50%   | 4.000000    | 7.000000    | 44.000000   | 16.000000           | 82.000000             | 71.000000                 |
| 75%   | 11.000000   | 13.000000   | 159.000000  | 16.000000           | 99.000000             | 82.000000                 |
| max   | 2640.000000 | 115.000000  | 6018.000000 | 546.000000          | 2753.000000           | 600.000000                |

|    | directoryPath                                                                                          | directoryParentPath                                                                            | directoryName              |
|----|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------|
| 0  | AxonFramework-5.0.2/.idea/copyright                                                                    | AxonFramework-5.0.2/.idea                                                                      | copyright                  |
| 1  | AxonFramework-5.0.2/.idea/inspectionProfiles                                                           | AxonFramework-5.0.2/.idea                                                                      | inspectionProfile          |
| 2  | AxonFramework-5.0.2/axon-framework-bom                                                                 | AxonFramework-5.0.2                                                                            | axon-framework-bom         |
| 3  | AxonFramework-5.0.2/axon-server-connector/src/main/resources/META-INF/services                         | AxonFramework-5.0.2/axon-server-connector/src/main                                             | resources/META-INF/service |
| 4  | AxonFramework-5.0.2/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/command | AxonFramework-5.0.2/axon-server-connector/src/test/java/org/axonframework/axonserver/connector | command                    |
| 5  | AxonFramework-5.0.2/axon-server-connector/src/test/resources                                           | AxonFramework-5.0.2/axon-server-connector/src/test                                             | resource                   |
| 6  | AxonFramework-5.0.2/build/coverage-report                                                              | AxonFramework-5.0.2/build                                                                      | coverage-report            |
| 7  | AxonFramework-5.0.2/build/parent                                                                       | AxonFramework-5.0.2/build                                                                      | parent                     |
| 8  | AxonFramework-5.0.2/common/src/main/java/org/axonframework/common/digest                               | AxonFramework-5.0.2/common/src/main/java/org/axonframework/common                              | digest                     |
| 9  | AxonFramework-5.0.2/common/src/main/java/org/axonframework/common/io                                   | AxonFramework-5.0.2/common/src/main/java/org/axonframework/common                              | io                         |
| 10 | AxonFramework-5.0.2/common/src/main/java/org/axonframework/common/tx                                   | AxonFramework-5.0.2/common/src/main/java/org/axonframework/common                              | tx                         |
| 11 | AxonFramework-5.0.2/common/src/main/resources/META-INF/services                                        | AxonFramework-5.0.2/common/src/main                                                            | resources/META-INF/service |
| 12 | AxonFramework-5.0.2/common/src/test/java/org/axonframework/common/lifecycle                            | AxonFramework-5.0.2/common/src/test/java/org/axonframework/common                              | lifecycle                  |
| 13 | AxonFramework-5.0.2/common/src/test/resources/META-INF/services                                        | AxonFramework-5.0.2/common/src/test                                                            | resources/META-INF/service |
| 14 | AxonFramework-5.0.2/conversion/src/main/resources/META-INF/services                                    | AxonFramework-5.0.2/conversion/src/main                                                        | resources/META-INF/service |
| 15 | AxonFramework-5.0.2/docs/_playbook/localLinks                                                          | AxonFramework-5.0.2/docs/_playbook                                                             | localLink                  |
| 16 | AxonFramework-5.0.2/docs/_reference/modules/ROOT/attachments                                           | AxonFramework-5.0.2/docs/_reference/modules/ROOT                                               | attachment                 |
| 17 | AxonFramework-5.0.2/docs/_reference/modules/ROOT/partials                                              | AxonFramework-5.0.2/docs/_reference/modules/ROOT                                               | partial                    |
| 18 | AxonFramework-5.0.2/docs/_reference/modules/axon-server-connector/pages                                | AxonFramework-5.0.2/docs/_reference/modules/axon-server-connector                              | page                       |
| 19 | AxonFramework-5.0.2/docs/_reference/modules/axon-server-connector/partials                             | AxonFramework-5.0.2/docs/_reference/modules/axon-server-connector                              | partial                    |
| 20 | AxonFramework-5.0.2/docs/_reference/modules/configuration/pages                                        | AxonFramework-5.0.2/docs/_reference/modules/configuration                                      | page                       |
| 21 | AxonFramework-5.0.2/docs/_reference/modules/configuration/partials                                     | AxonFramework-5.0.2/docs/_reference/modules/configuration                                      | partial                    |
| 22 | AxonFramework-5.0.2/docs/_reference/modules/disruptor/pages                                            | AxonFramework-5.0.2/docs/_reference/modules/disruptor                                          | page                       |
| 23 | AxonFramework-5.0.2/docs/_reference/modules/disruptor/partials                                         | AxonFramework-5.0.2/docs/_reference/modules/disruptor                                          | partial                    |
| 24 | AxonFramework-5.0.2/docs/_reference/modules/eventsourcing/pages                                        | AxonFramework-5.0.2/docs/_reference/modules/eventsourcing                                      | page                       |
| 25 | AxonFramework-5.0.2/docs/_reference/modules/eventsourcing/partials                                     | AxonFramework-5.0.2/docs/_reference/modules/eventsourcing                                      | partial                    |
| 26 | AxonFramework-5.0.2/docs/_reference/modules/legacy/pages                                               | AxonFramework-5.0.2/docs/_reference/modules/legacy                                             | page                       |
| 27 | AxonFramework-5.0.2/docs/_reference/modules/legacy/partials                                            | AxonFramework-5.0.2/docs/_reference/modules/legacy                                             | partial                    |
| 28 | AxonFramework-5.0.2/docs/_reference/modules/metrics/pages                                              | AxonFramework-5.0.2/docs/_reference/modules/metrics                                            | page                       |
| 29 | AxonFramework-5.0.2/docs/_reference/modules/metrics/partials                                           | AxonFramework-5.0.2/docs/_reference/modules/metrics                                            | partial                    |

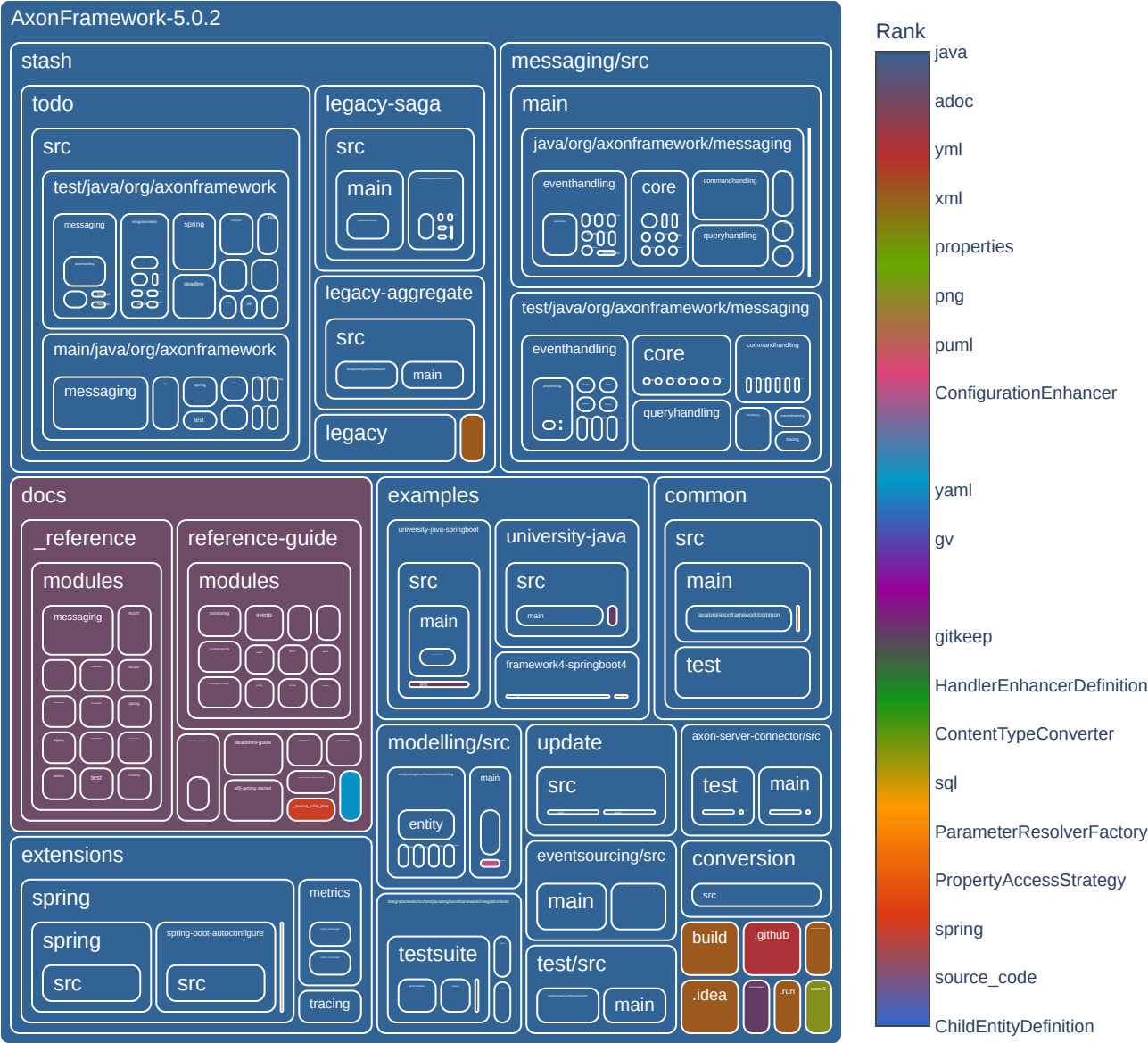
## Number of files per directory

## Directories and their file count



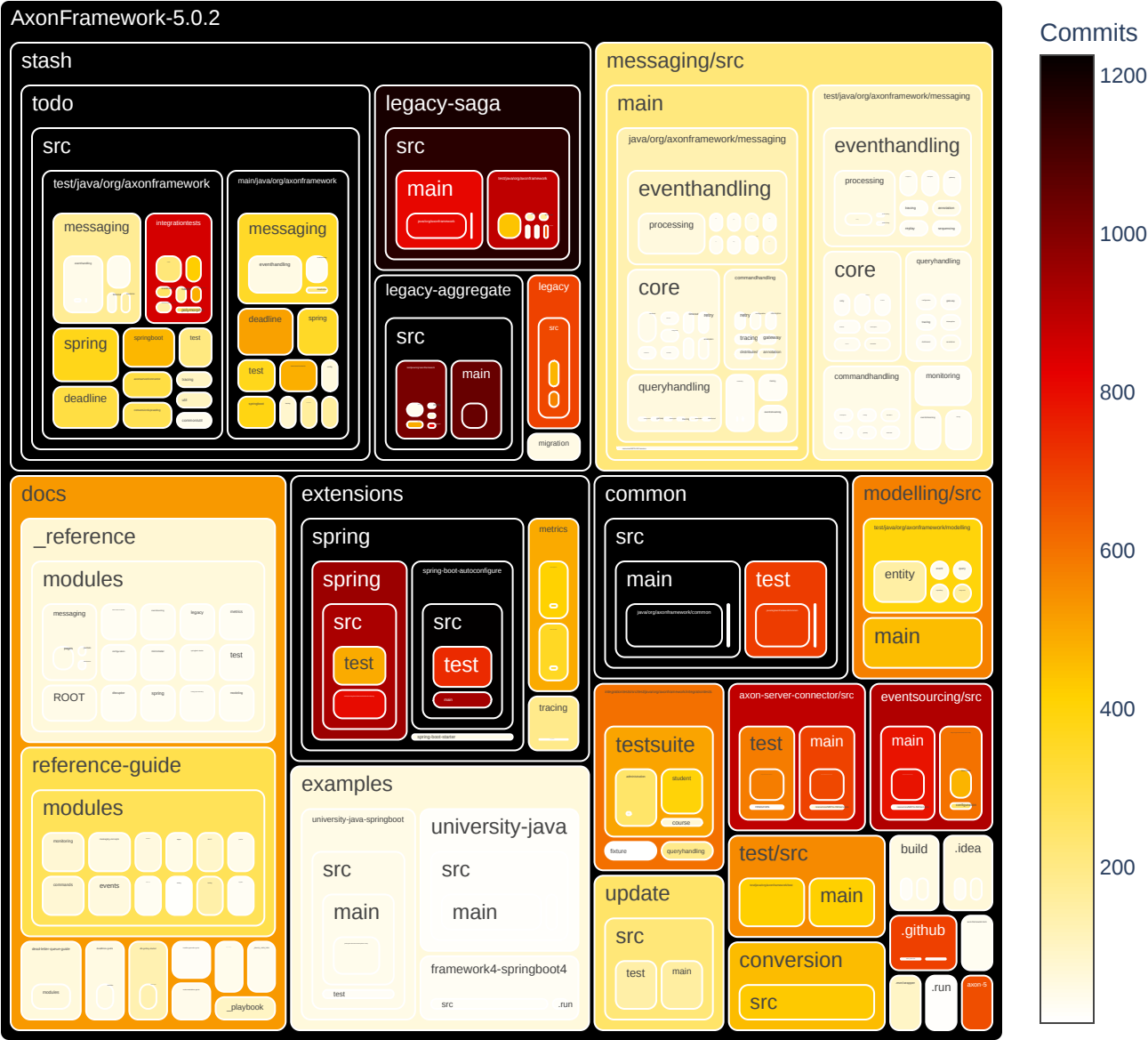
## Most frequent file extension per directory

Most frequent file extension per directory



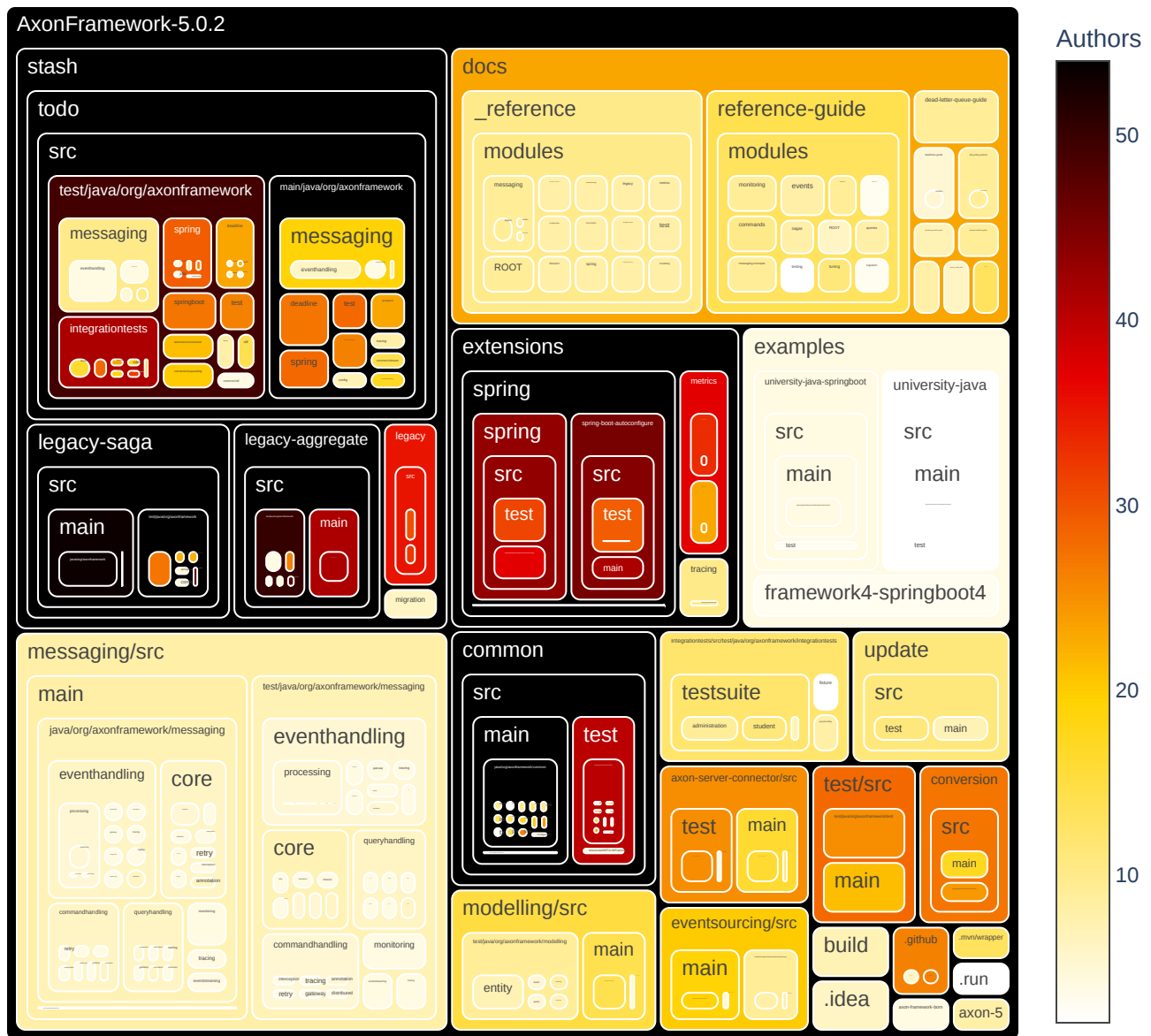
Number of commits per directory

Number of git commits



Number of distinct authors per directory

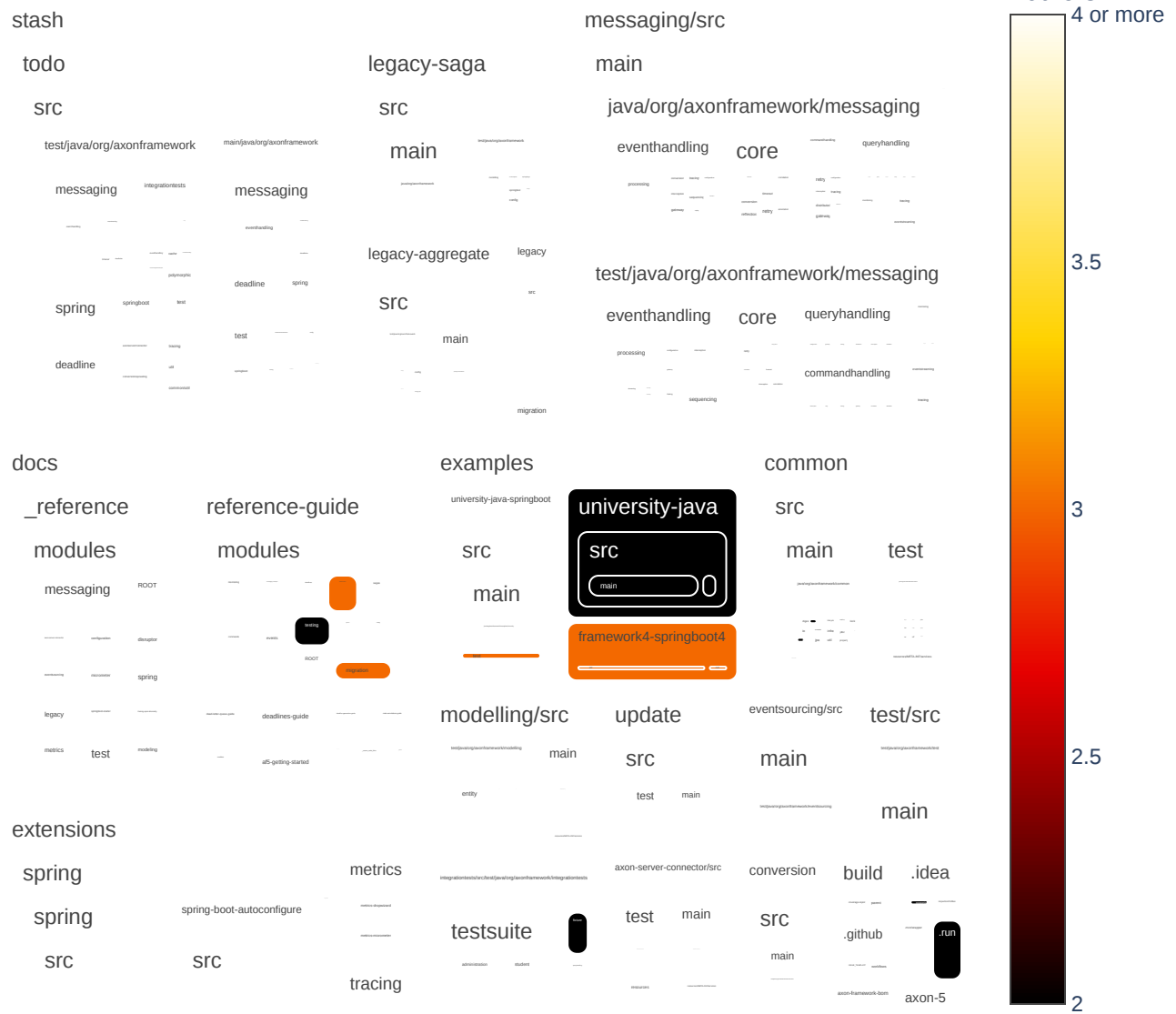
## Number of distinct commit authors



## Directories with very few different authors

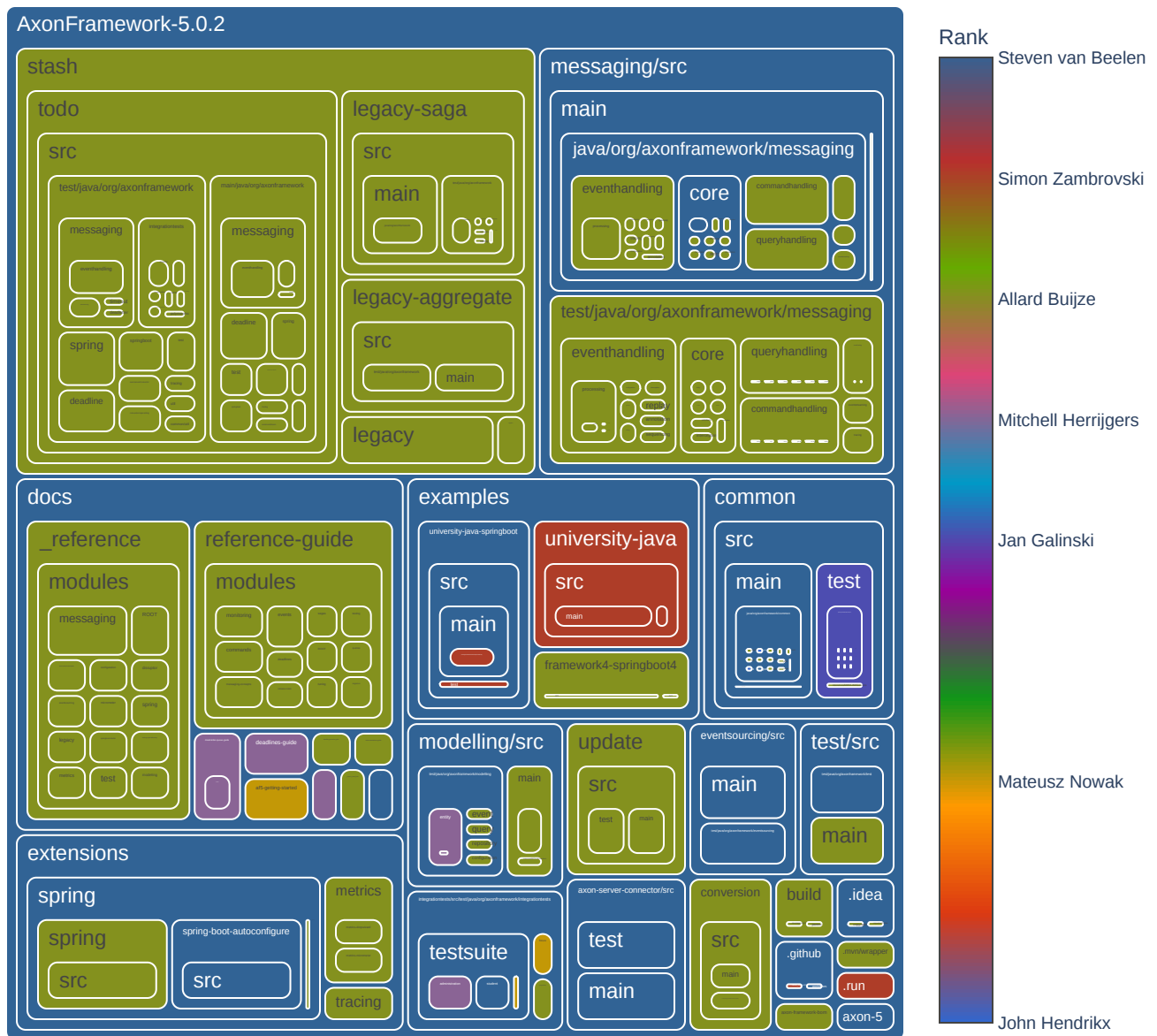
Number of distinct commit authors (red/black = only one or very few authors)

AxonFramework-5.0.2



Main author per directory

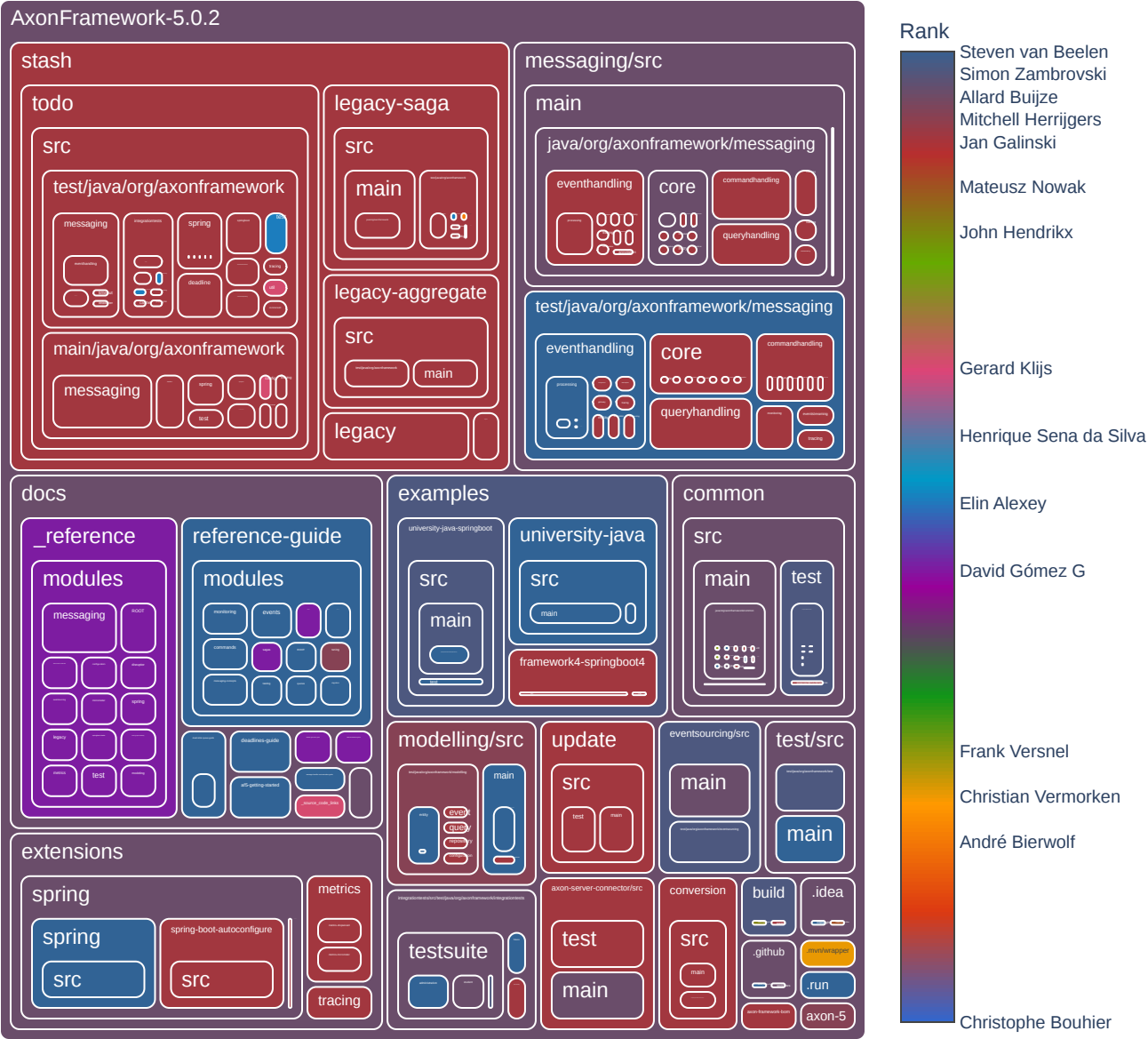
## Main authors with highest number of commits



## Second author per directory



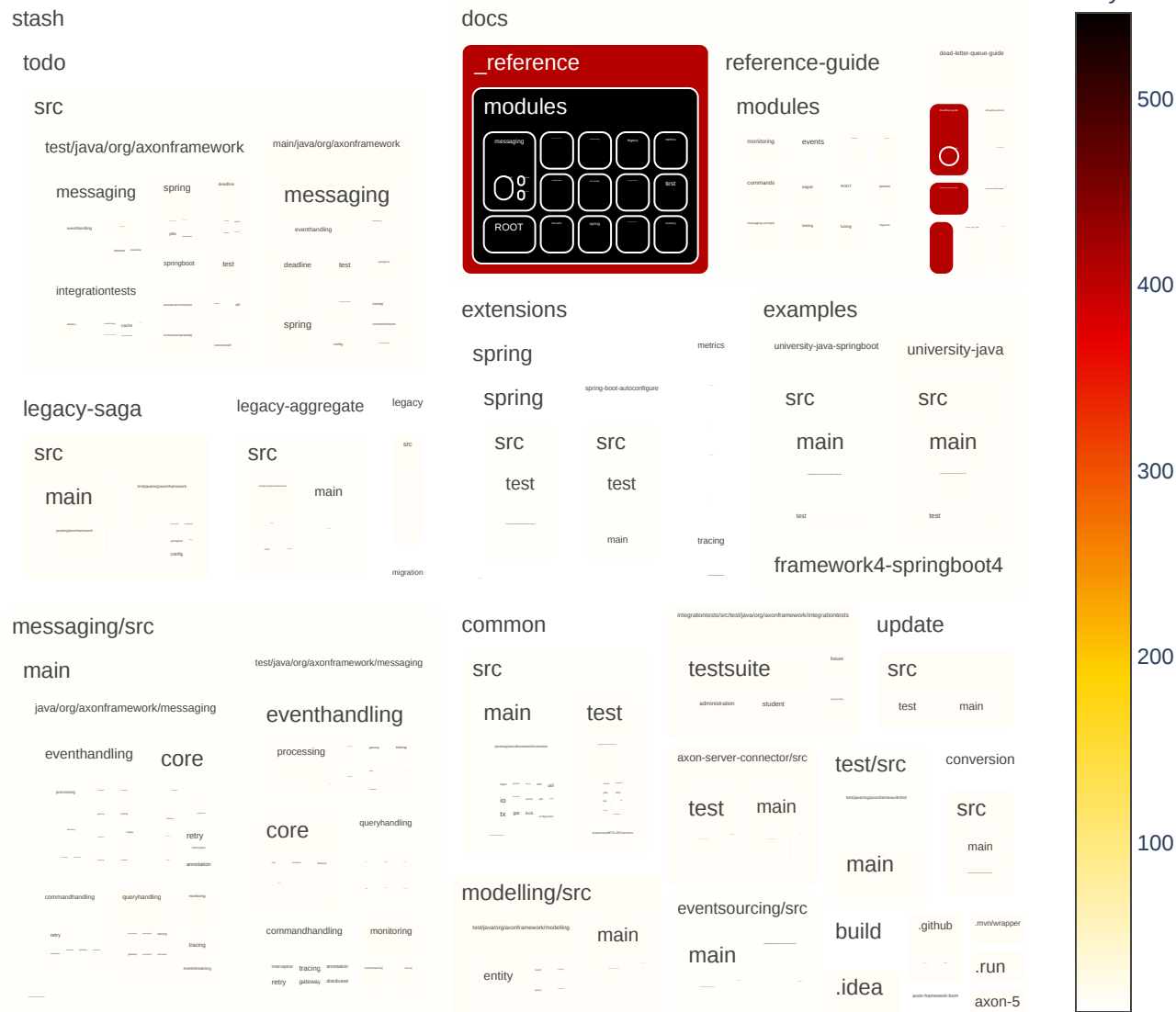
Second author with the second highest number of commits



Days since last commit per directory

Days since last commit

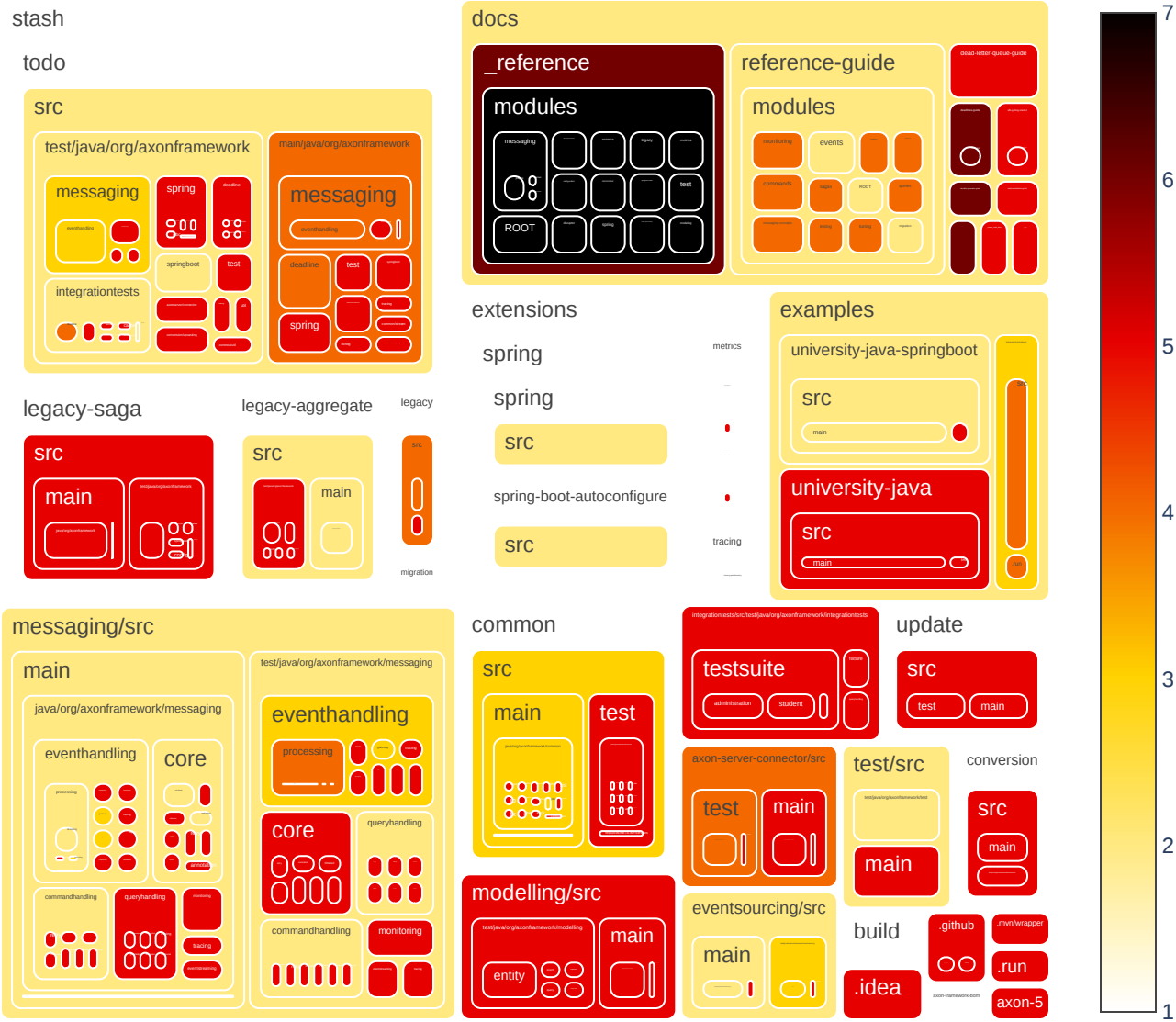
AxonFramework-5.0.2



Days since last commit per directory (ranked)

Rank of days since last commit

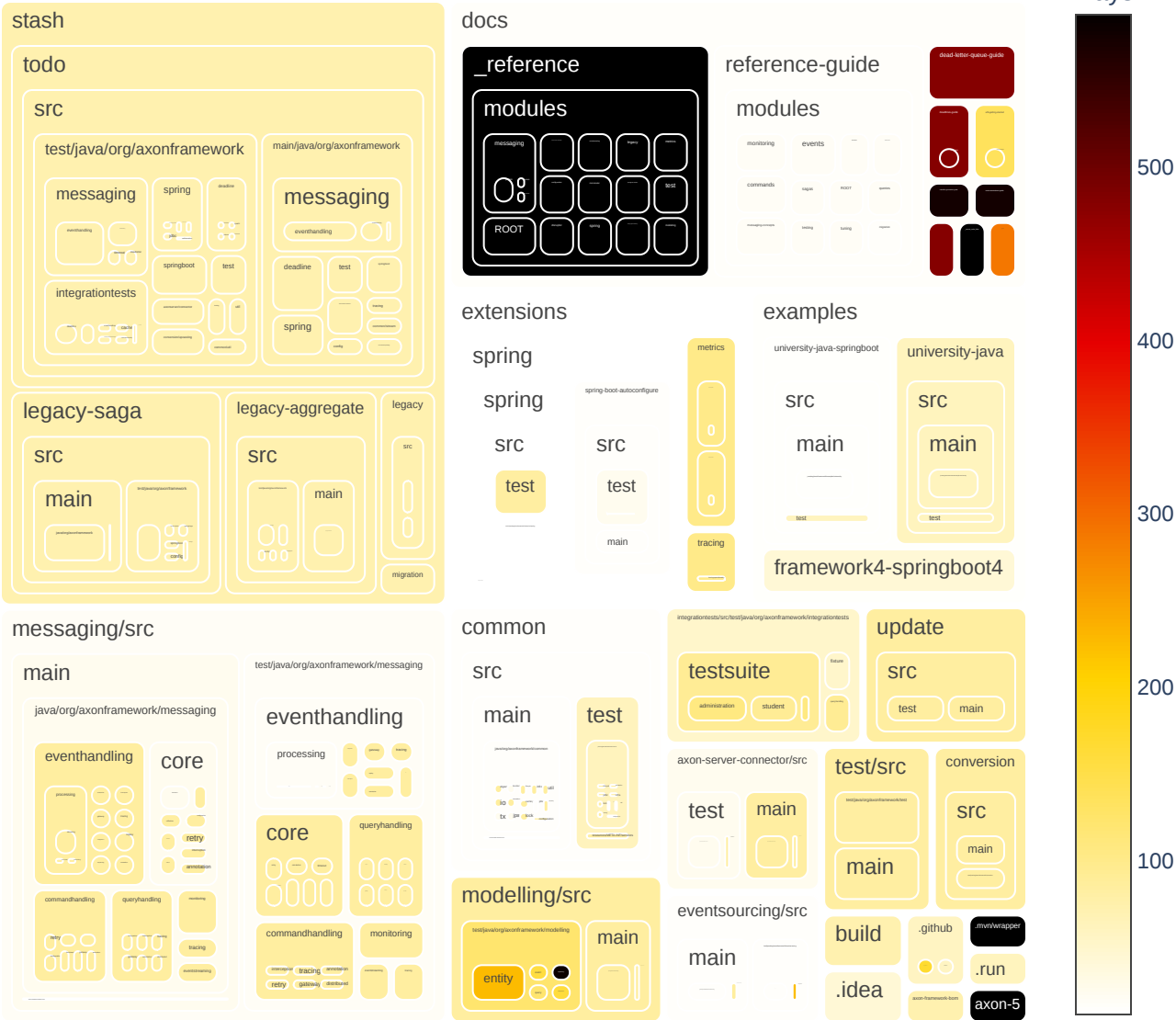
AxonFramework-5.0.2



Days since last file creation per directory

Days since last file creation

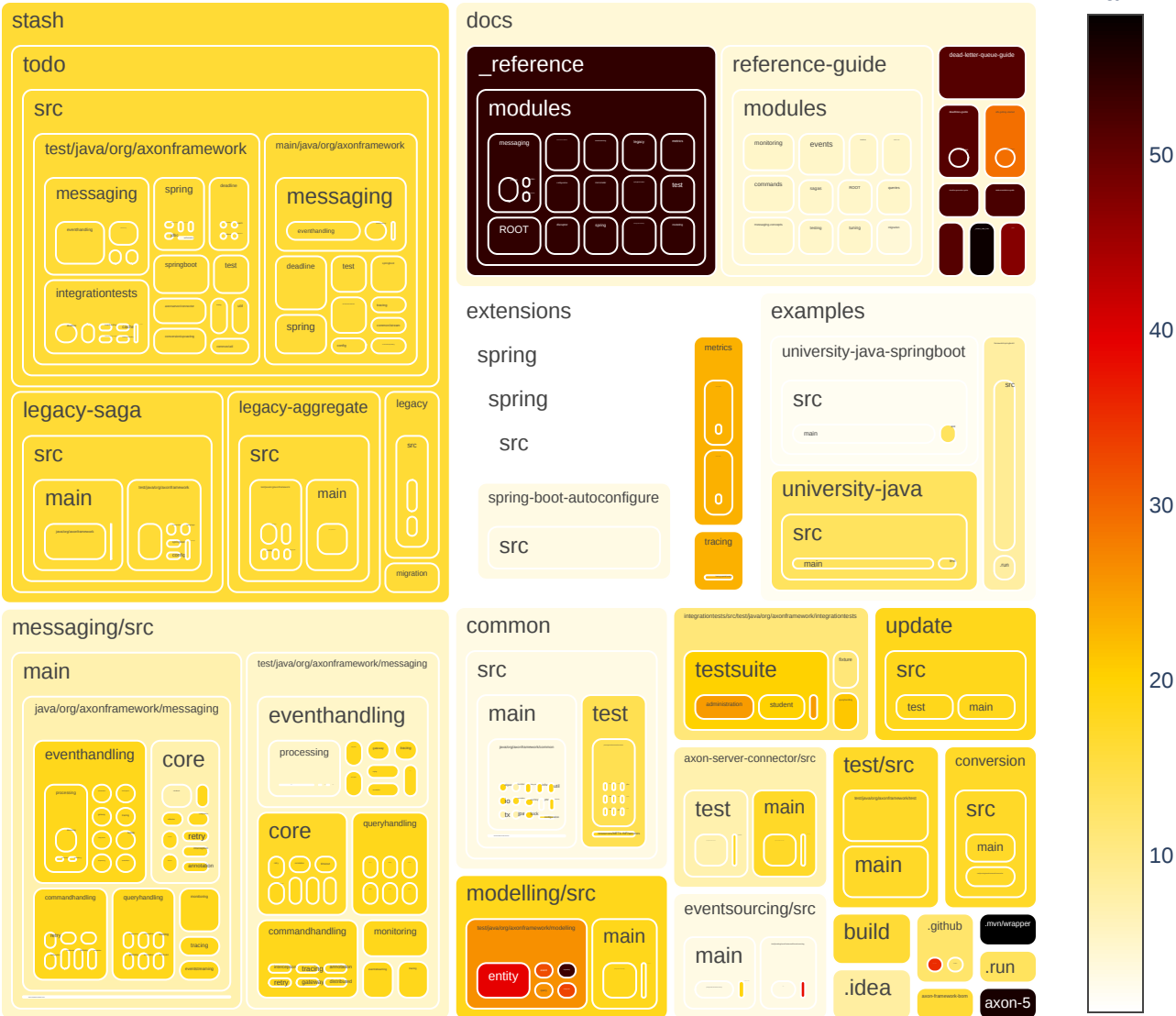
AxonFramework-5.0.2



Days since last file creation per directory (ranked)

Rank of days since last file creation

AxonFramework-5.0.2



Days since last file modification per directory

## Days since last file modification

AxonFramework-5.0.2

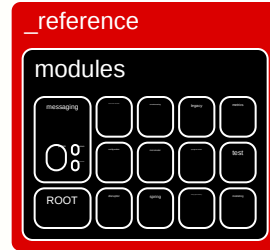
stash

todo

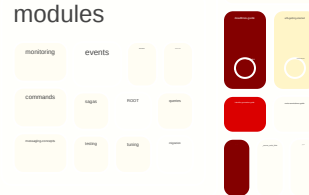
src



docs



reference-guide



extensions

spring

spring

src

test

spring

src

test

main

spring

src

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main

examples

university-java-springboot

university-java

src

main

test

framework4-springboot4

legacy-saga

src

main

legacy-aggregate

src

main

legacy

src

main

migration

messaging/src

main

java/org/axonframework/messaging

eventhandling

core

eventhandling

core

eventhandling

core

test/java/org/axonframework/messaging

eventhandling

processing

core

core

eventhandling

core

eventhandling

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src

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integrationtests/src/test/java/org/axonframework/integrationtests

testsuite

administration

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Days



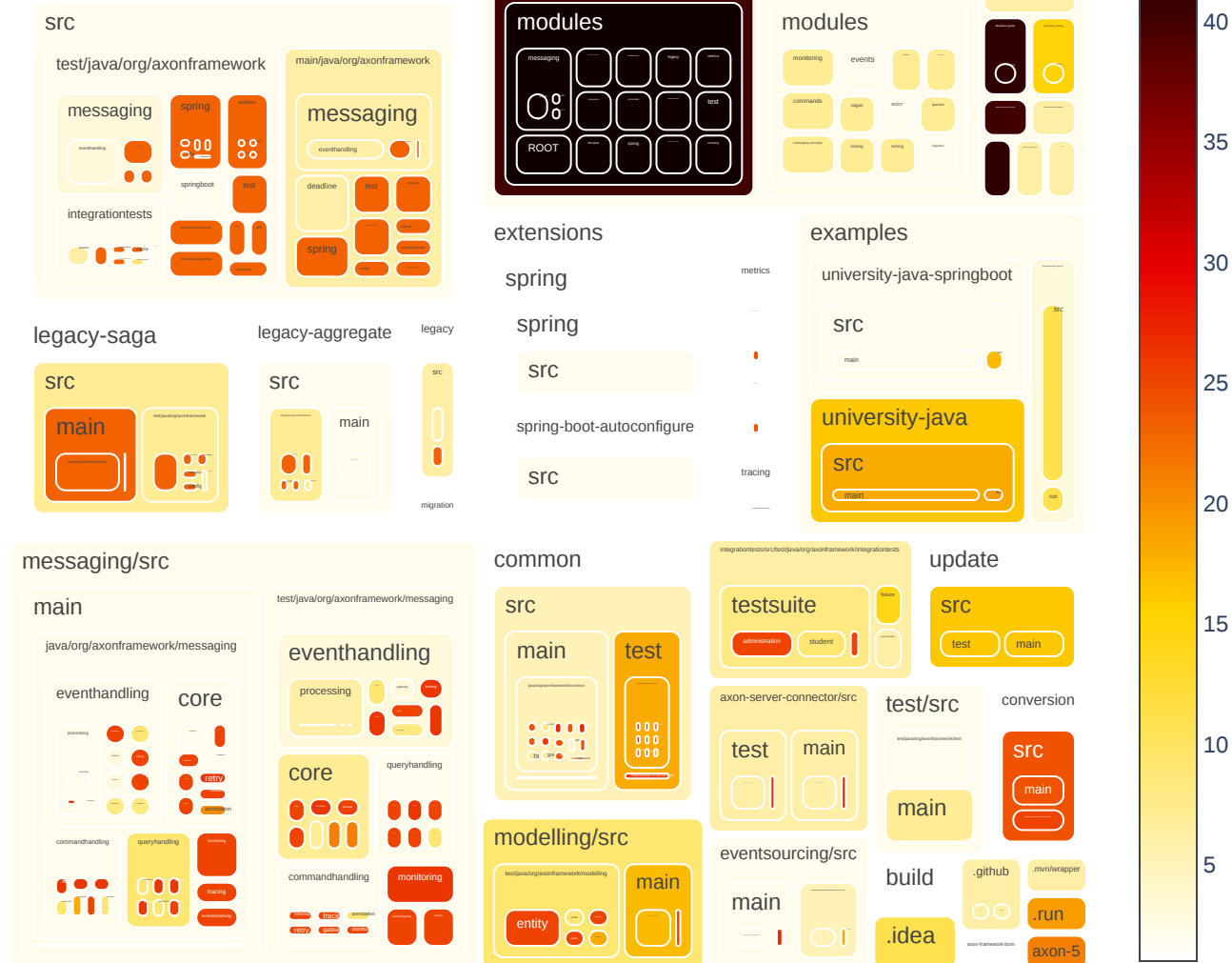
Days since last file modification per directory (ranked)

## Rank of days since last file modification

AxonFramework-5.0.2

stash

todo



## Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

## Preview data

Sum of commits that changed more than 30 files (each) = 1679

Max changed files with one commit = 4781

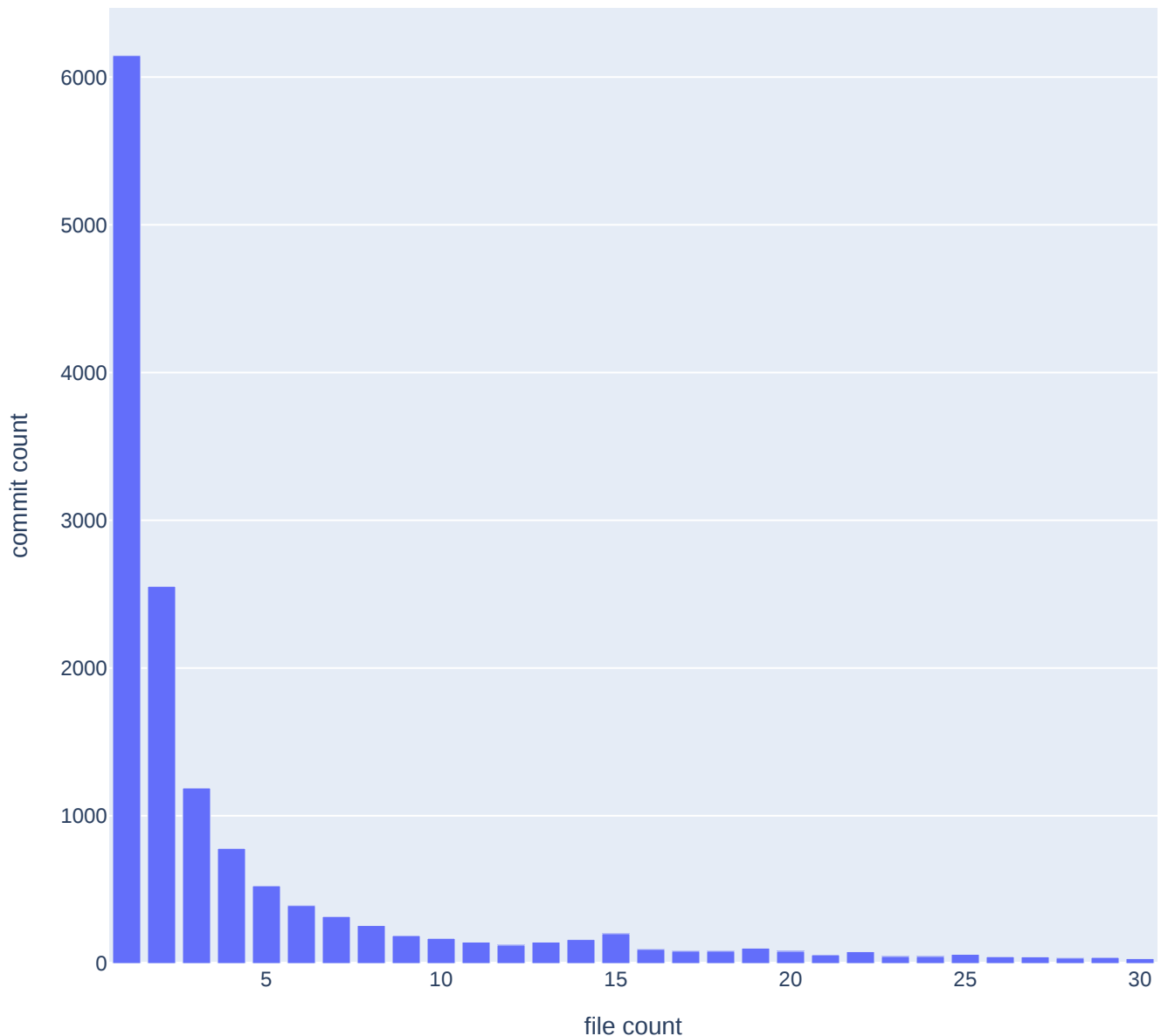
|       | filesPerCommit | commitCount |
|-------|----------------|-------------|
| count | 475.000000     | 475.000000  |
| mean  | 572.324211     | 33.526316   |
| std   | 826.005356     | 313.673903  |
| min   | 1.000000       | 1.000000    |
| 25%   | 119.500000     | 1.000000    |
| 50%   | 260.000000     | 2.000000    |
| 75%   | 593.000000     | 5.000000    |
| max   | 4781.000000    | 6145.000000 |

|    | filesPerCommit | commitCount |
|----|----------------|-------------|
| 0  | 1              | 6145        |
| 1  | 2              | 2553        |
| 2  | 3              | 1188        |
| 3  | 4              | 779         |
| 4  | 5              | 526         |
| 5  | 6              | 392         |
| 6  | 7              | 318         |
| 7  | 8              | 257         |
| 8  | 9              | 189         |
| 9  | 10             | 170         |
| 10 | 11             | 145         |
| 11 | 12             | 125         |
| 12 | 13             | 145         |
| 13 | 14             | 162         |
| 14 | 15             | 202         |
| 15 | 16             | 96          |
| 16 | 17             | 84          |
| 17 | 18             | 84          |
| 18 | 19             | 104         |
| 19 | 20             | 83          |
| 20 | 21             | 59          |
| 21 | 22             | 80          |
| 22 | 23             | 48          |
| 23 | 24             | 48          |
| 24 | 25             | 62          |
| 25 | 26             | 46          |
| 26 | 27             | 45          |
| 27 | 28             | 37          |
| 28 | 29             | 41          |
| 29 | 30             | 33          |

Bar chart with the number of files per commit distribution



## Changed files per commit



## Pairwise Changed Files

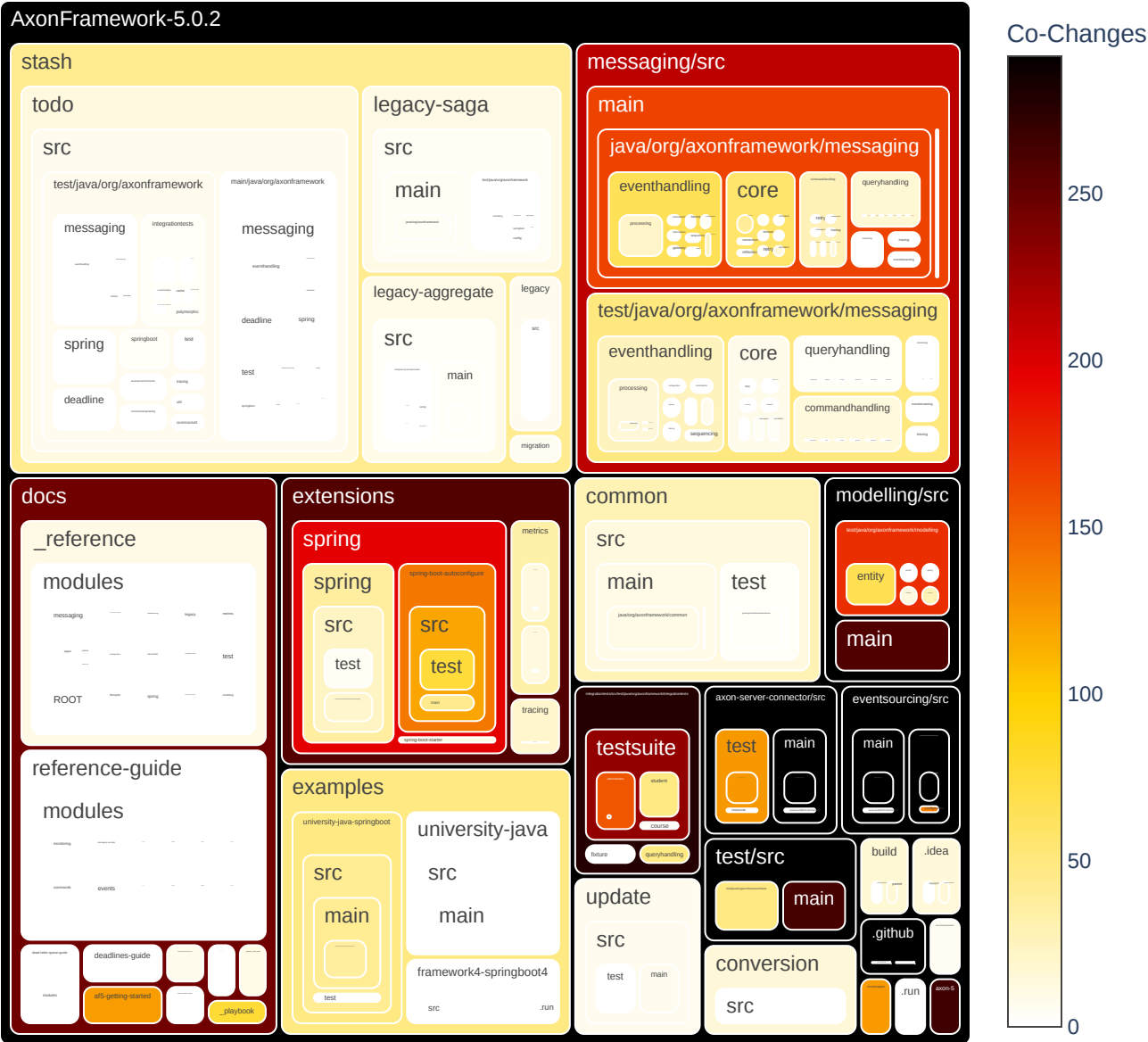
This section analyzes files that were changed together within the same commit and provides several metrics to quantify the strength of the co-change relationship:

- **Commit Count:** The number of commits in which two files were changed together.
- **Commit Lift:** A ratio that indicates whether the co-change pattern is stronger than random chance, given how often each file changes.
- **Jaccard Similarity:** The ratio of commits involving either file that also involved both files.

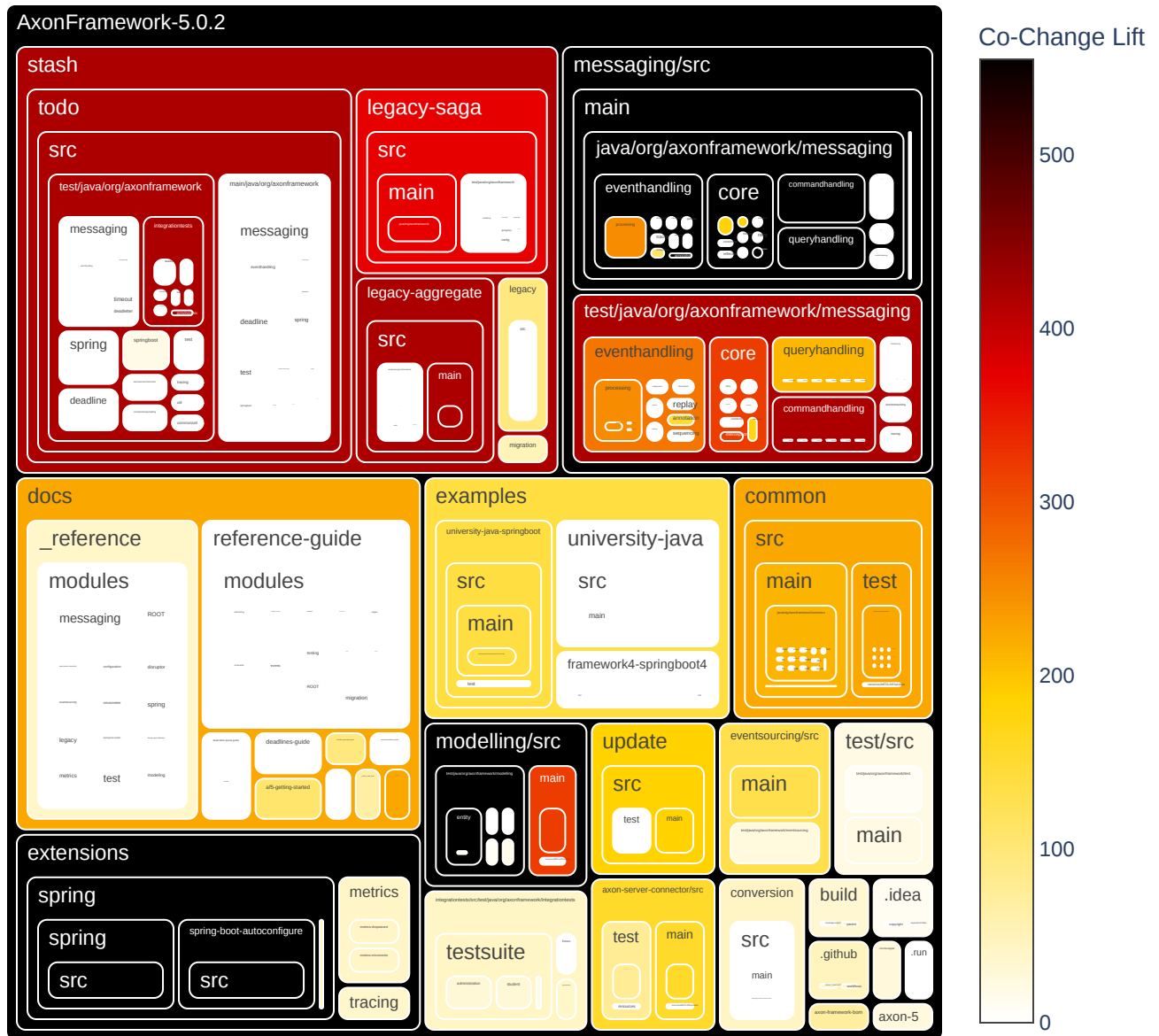
The following tables show the top pairwise changed files based on these metrics. The following charts show how these metrics are distributed across pairs of files that were changed together.

## Treemap with files changed frequently with others

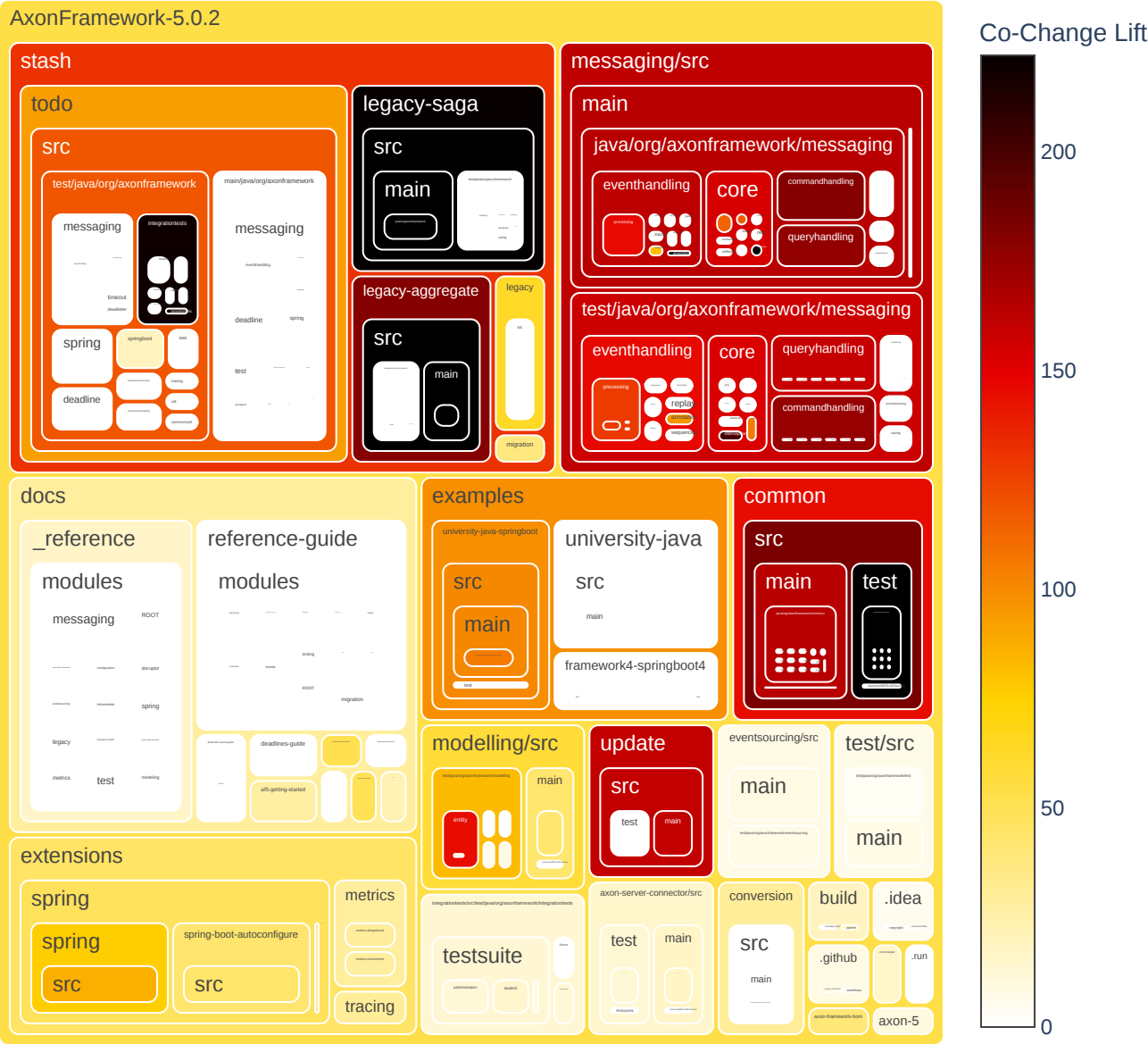
Files that likely co-change with others in update commits



Co-Changing files in update commits max lift (1=random, >1=more than random, <1=



Co-Changing files in update commits average lift (1=random, >1=more than random,

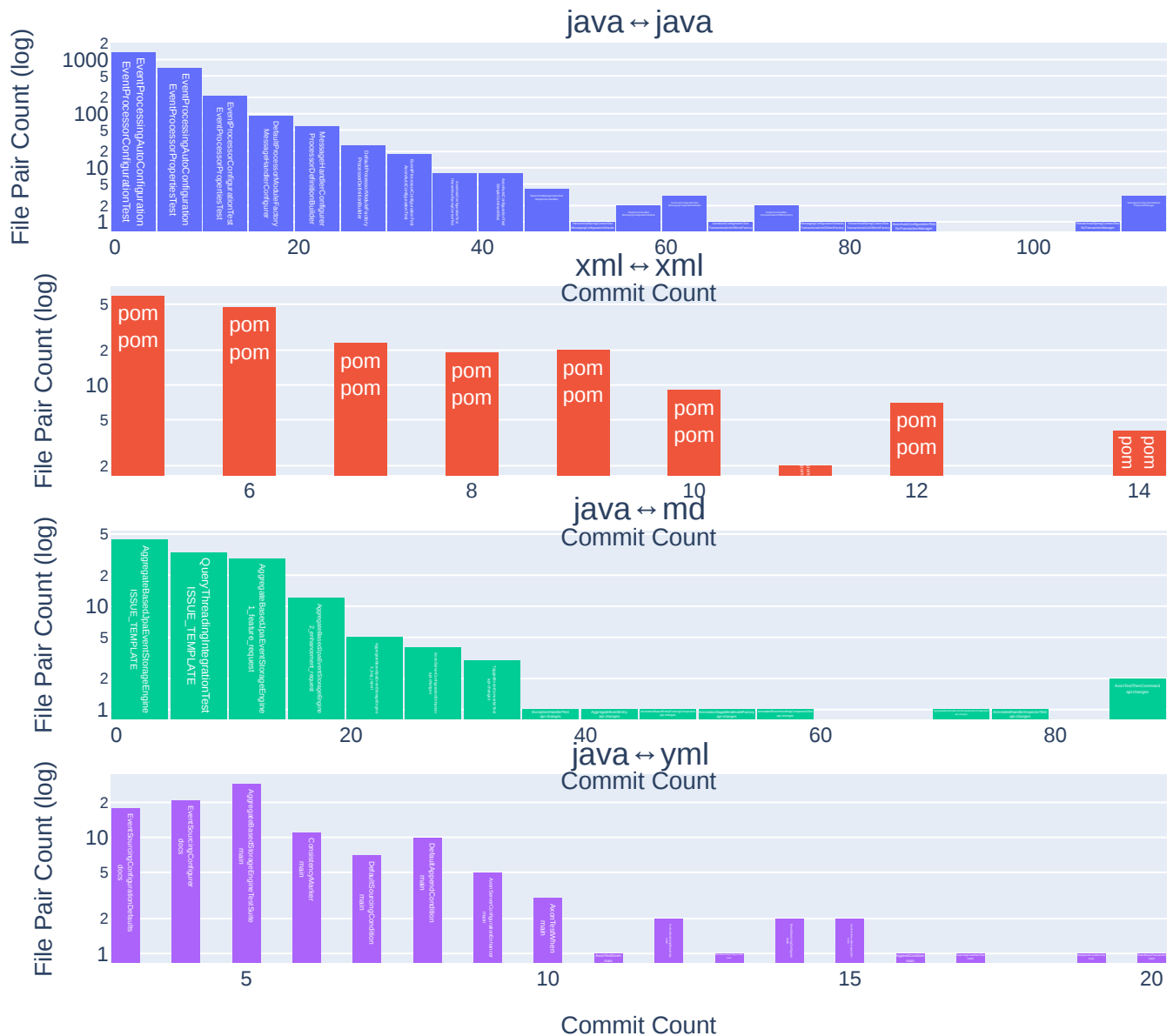


|   | fileExtensionPair | fileExtensionPairCount |
|---|-------------------|------------------------|
| 0 | java↔java         | 2521                   |
| 1 | xml↔xml           | 190                    |
| 2 | java↔md           | 139                    |
| 3 | java↔yaml         | 116                    |

Files changed together by commit count

|    | fileExtensionPair | updateCommitCount | GroupRank | filePair                                                                                 |
|----|-------------------|-------------------|-----------|------------------------------------------------------------------------------------------|
| 0  | java↔java         | 114               | 1         | StorageEngineTestSuite↔AggregateBasedStorageEngineTestSuite                              |
| 1  | java↔java         | 112               | 2         | AggregateBasedAxonServerEventStorageEngine↔AggregateBasedJpaEventStorageEngine connector |
| 2  | java↔java         | 111               | 3         | AggregateBasedJpaEventStorageEngine↔AggregateBasedStorageEngineTestSuite ev              |
| 3  | java↔java         | 109               | 4         | AggregateBasedAxonServerEventStorageEngine↔AggregateBasedStorageEngineTestSuite conn     |
| 4  | java↔java         | 85                | 5         | StorageEngineTestSuite↔AggregateBasedJpaEventStorageEngine                               |
| 5  | java↔java         | 83                | 6         | StorageEngineTestSuite↔AggregateBasedAxonServerEventStorageEngine e                      |
| 6  | java↔java         | 76                | 7         | StorageEngineTestSuite↔DefaultEventStoreTransactionTest                                  |
| 7  | java↔java         | 73                | 8         | AggregateBasedStorageEngineTestSuite↔DefaultEventStoreTransactionTest                    |
| 8  | java↔java         | 72                | 9         | StorageEngineTestSuite↔DefaultEventStoreTransaction                                      |
| 9  | java↔java         | 65                | 10        | DefaultEventStoreTransaction↔AggregateBasedStorageEngineTestSuite                        |
| 10 | xml↔xml           | 14                | 1         | pom↔pom                                                                                  |
| 11 | xml↔xml           | 12                | 2         | pom↔pom                                                                                  |
| 12 | xml↔xml           | 11                | 3         | pom↔pom                                                                                  |
| 13 | xml↔xml           | 10                | 4         | pom↔pom                                                                                  |
| 14 | xml↔xml           | 9                 | 5         | pom↔pom                                                                                  |
| 15 | xml↔xml           | 8                 | 6         | pom↔pom                                                                                  |
| 16 | xml↔xml           | 7                 | 7         | pom↔pom                                                                                  |
| 17 | xml↔xml           | 6                 | 8         | pom↔pom                                                                                  |
| 18 | xml↔xml           | 5                 | 9         | pom↔pom                                                                                  |
| 19 | java↔md           | 87                | 1         | AggregateBasedStorageEngineTestSuite↔api-changes                                         |
| 20 | java↔md           | 86                | 2         | AggregateBasedAxonServerEventStorageEngine↔api-changes                                   |
| 21 | java↔md           | 78                | 3         | AggregateBasedJpaEventStorageEngine↔api-changes                                          |
| 22 | java↔md           | 70                | 4         | StorageEngineTestSuite↔api-changes                                                       |
| 23 | java↔md           | 55                | 5         | SimpleEventStoreTest↔api-changes                                                         |
| 24 | java↔md           | 51                | 6         | DefaultEventStoreTransactionTest↔api-changes                                             |
| 25 | java↔md           | 46                | 7         | QueryThreadingIntegrationTest↔api-changes                                                |
| 26 | java↔md           | 41                | 8         | DefaultEventStoreTransaction↔api-changes                                                 |
| 27 | java↔md           | 39                | 9         | EventSourcingConfigurationDefaults↔api-changes                                           |
| 28 | java↔md           | 34                | 10        | AxonServerEventStorageEngine↔api-changes                                                 |
| 29 | java↔yaml         | 20                | 1         | DefaultEventStoreTransaction↔pullrequest                                                 |
| 30 | java↔yaml         | 19                | 2         | StorageEngineTestSuite↔pullrequest                                                       |
| 31 | java↔yaml         | 17                | 3         | AggregateBasedStorageEngineTestSuite↔pullrequest                                         |
| 32 | java↔yaml         | 16                | 4         | AggregateBasedStorageEngineTestSuite↔main                                                |
| 33 | java↔yaml         | 15                | 5         | StorageEngineTestSuite↔main                                                              |
| 34 | java↔yaml         | 14                | 6         | DefaultEventStoreTransaction↔main                                                        |
| 35 | java↔yaml         | 13                | 7         | QueryThreadingIntegrationTest↔pullrequest                                                |
| 36 | java↔yaml         | 12                | 8         | QueryThreadingIntegrationTest↔main                                                       |
| 37 | java↔yaml         | 11                | 9         | AppendCondition↔pullrequest                                                              |
| 38 | java↔yaml         | 10                | 10        | NoAppendCondition↔main                                                                   |

## Co-Changed Files by their commit count

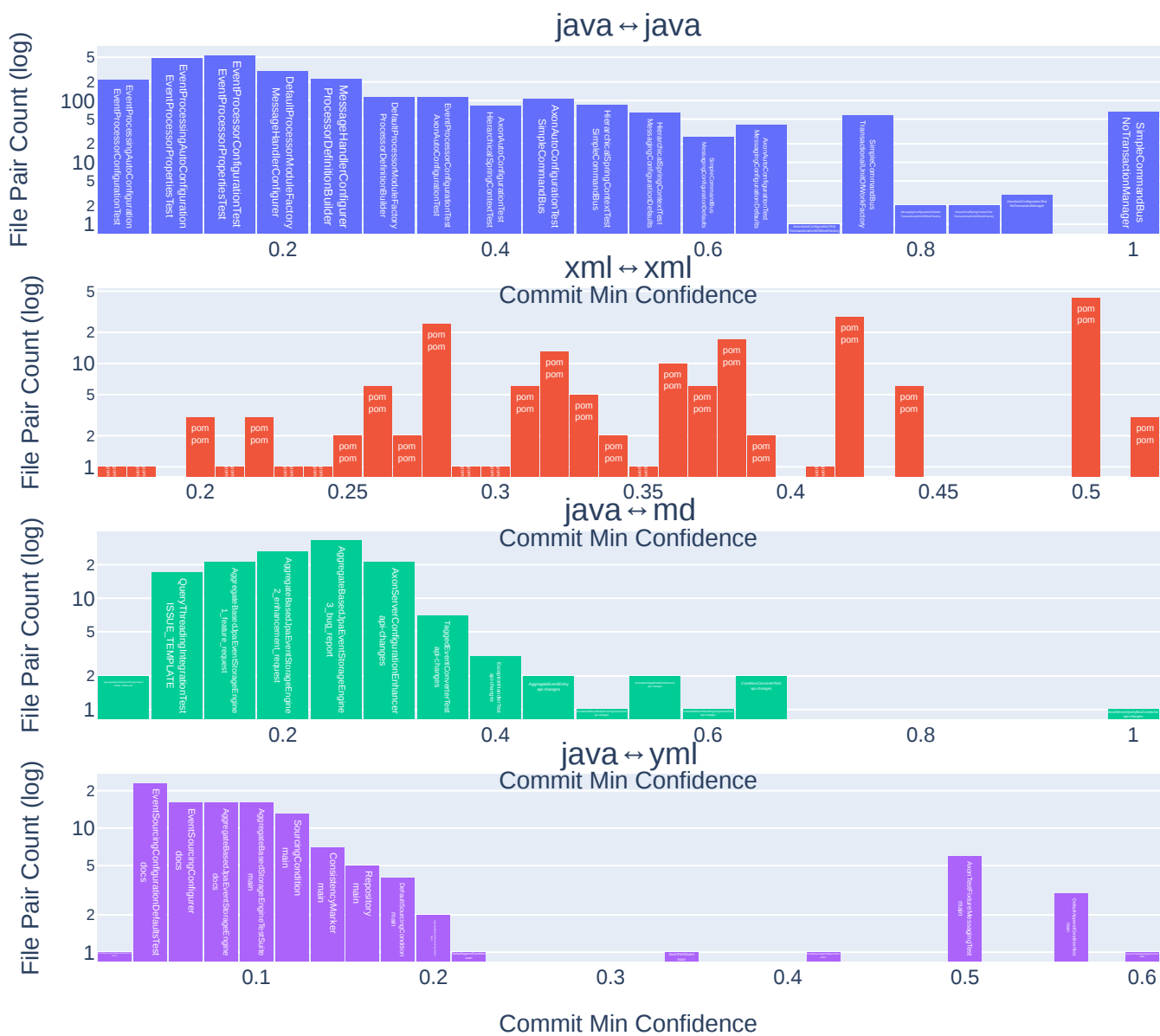


## Files changed together by commit min confidence

The commit min confidence is the commit count where both files were changed divided by the commit count of the file with the least commits. This metric is useful to identify pairs of files that are frequently changed together and is not biased by single files that are changed very often.

|    | fileExtensionPair | updateCommitMinConfidence | GroupRank | filePair                                                          |                    |
|----|-------------------|---------------------------|-----------|-------------------------------------------------------------------|--------------------|
| 0  | java↔java         | 1.000000                  | 1         | DefaultProcessorModuleFactory↔MessageHandlerConfigurer            | extensio           |
| 1  | java↔java         | 0.900000                  | 2         | ChildEntityFieldDefinition↔GetterSetterChildEntityFieldDefinition |                    |
| 2  | java↔java         | 0.875000                  | 3         | ChildEntityFieldDefinition↔FieldChildEntityFieldDefinition        |                    |
| 3  | java↔java         | 0.833333                  | 4         | AnnotatedEventHandlingComponent↔AnnotationEventHandlerAdapter     | messaging/src/mail |
| 4  | java↔java         | 0.800000                  | 5         | EventProcessorInfoUtils↔SpringUtils                               |                    |
| 5  | java↔java         | 0.750000                  | 6         | TransactionalUnitOfWorkFactory↔CommandBusTestUtils                |                    |
| 6  | java↔java         | 0.714286                  | 7         | FutureUtils↔DefaultAxonApplication                                |                    |
| 7  | java↔java         | 0.666667                  | 8         | FutureUtils↔DefaultAxonApplicationTest                            |                    |
| 8  | java↔java         | 0.661538                  | 9         | DefaultSourcingConditionTest↔SourcingConditionTest                |                    |
| 9  | java↔java         | 0.648148                  | 10        | DefaultAppendConditionTest↔NoAppendConditionTest                  |                    |
| 10 | xml↔xml           | 0.518519                  | 1         |                                                                   | pom↔pom            |
| 11 | xml↔xml           | 0.500000                  | 2         |                                                                   | pom↔pom            |
| 12 | xml↔xml           | 0.444444                  | 3         |                                                                   | pom↔pom            |
| 13 | xml↔xml           | 0.416667                  | 4         |                                                                   | pom↔pom            |
| 14 | xml↔xml           | 0.407407                  | 5         |                                                                   | pom↔pom            |
| 15 | xml↔xml           | 0.392857                  | 6         |                                                                   | pom↔pom            |
| 16 | xml↔xml           | 0.388889                  | 7         |                                                                   | pom↔pom            |
| 17 | xml↔xml           | 0.384615                  | 8         |                                                                   | pom↔pom            |
| 18 | xml↔xml           | 0.370370                  | 9         |                                                                   | pom↔pom            |
| 19 | xml↔xml           | 0.363636                  | 10        |                                                                   | pom↔pom            |
| 20 | java↔md           | 1.000000                  | 1         | AnnotationSagaMetaModelFactory↔api-changes                        |                    |
| 21 | java↔md           | 0.666667                  | 2         | AnnotationEventHandlerAdapter↔api-changes                         |                    |
| 22 | java↔md           | 0.600000                  | 3         | CommandConverter↔modularity                                       |                    |
| 23 | java↔md           | 0.571429                  | 4         | ExceptionHandlerTest↔api-changes                                  |                    |
| 24 | java↔md           | 0.500000                  | 5         | SimpleCommandHandlingComponentTest↔api-changes                    |                    |
| 25 | java↔md           | 0.445596                  | 6         | AggregateBasedAxonServerEventStorageEngine↔api-changes            |                    |
| 26 | java↔md           | 0.441176                  | 7         | ChildAmbiguityException↔api-changes                               |                    |
| 27 | java↔md           | 0.391304                  | 8         | Repository↔api-changes                                            |                    |
| 28 | java↔md           | 0.375000                  | 9         | GetterEvolverChildEntityFieldDefinition↔api-changes               |                    |
| 29 | java↔md           | 0.367925                  | 10        | AggregateBasedJpaEventStorageEngine↔api-changes                   |                    |
| 30 | java↔yaml         | 0.600000                  | 1         | SpringUtils↔slack-release-notification                            |                    |
| 31 | java↔yaml         | 0.555556                  | 2         | RenameCourseController↔application                                |                    |
| 32 | java↔yaml         | 0.500000                  | 3         | SubscribeStudentToCourseController↔application                    | ex                 |
| 33 | java↔yaml         | 0.428571                  | 4         | EventProcessingAutoConfiguration↔test-docker-compose              |                    |
| 34 | java↔yaml         | 0.333333                  | 5         | UpdateCheckerProperties↔application                               |                    |
| 35 | java↔yaml         | 0.222222                  | 6         | NoAppendCondition↔pullrequest                                     |                    |
| 36 | java↔yaml         | 0.203390                  | 7         | NoAppendConditionTest↔pullrequest                                 |                    |
| 37 | java↔yaml         | 0.193548                  | 8         | DefaultEventStoreTransaction↔add-to-project                       |                    |
| 38 | java↔yaml         | 0.185185                  | 9         | DefaultAppendConditionTest↔pullrequest                            |                    |
| 39 | java↔yaml         | 0.183333                  | 10        | AppendCondition↔pullrequest                                       |                    |

# Co-Changed Files by their commit min confidence

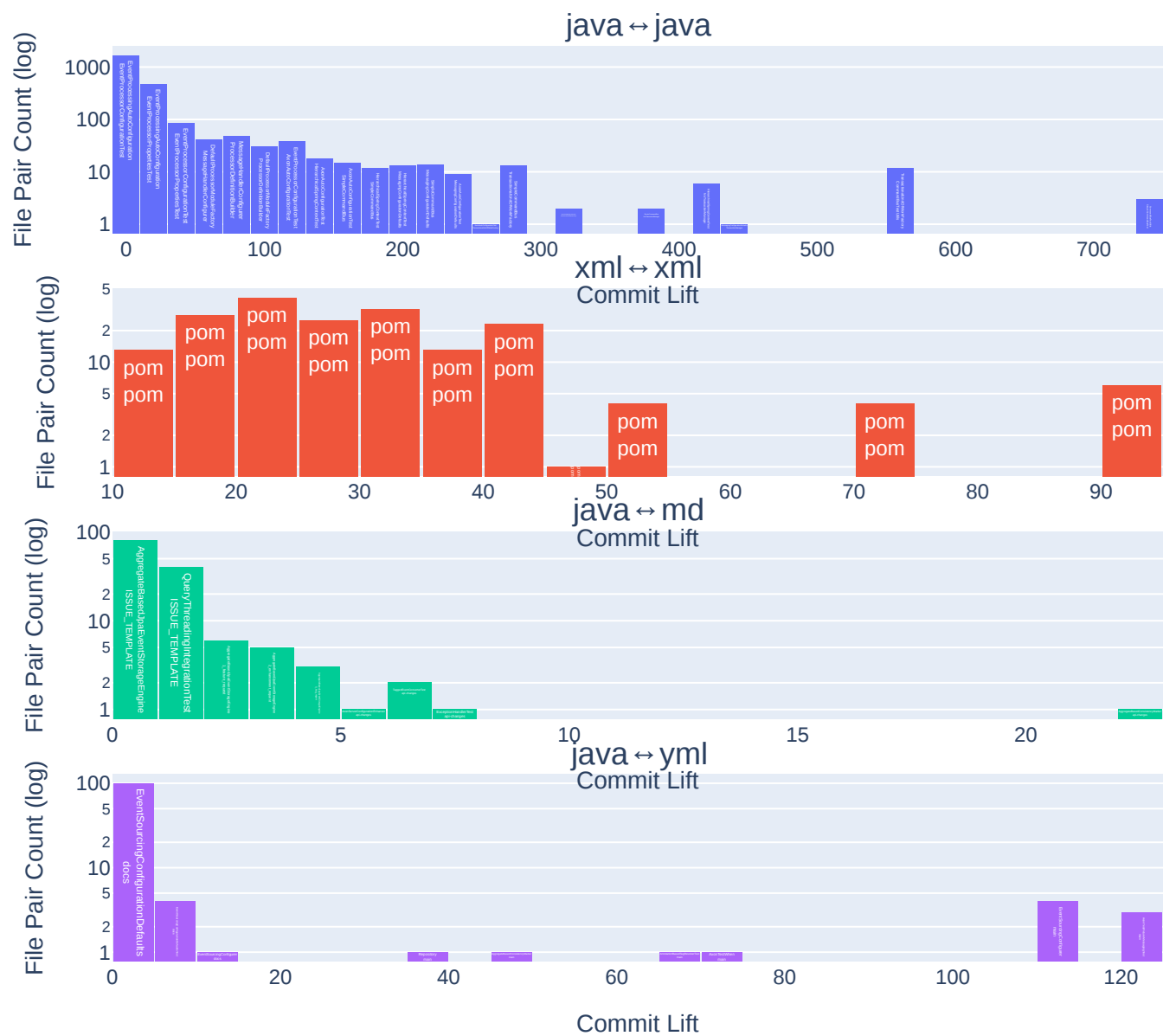


Files changed together by commit lift



|    | fileExtensionPair | updateCommitLift | GroupRank | filePair                                                                                      |
|----|-------------------|------------------|-----------|-----------------------------------------------------------------------------------------------|
| 0  | java↔java         | 740.000000       | 1         | FieldChildEntityFieldDefinitionTest↔GetterEvolverChildEntityFieldDefinitionTest               |
| 1  | java↔java         | 555.000000       | 2         | Command↔Event                                                                                 |
| 2  | java↔java         | 555.000000       | 3         | DefaultProcessorModuleFactory↔ProcessorDefinitionBuilder                                      |
| 3  | java↔java         | 444.000000       | 4         | MethodSequencingPolicyEventHandlerDefinition↔MethodSequencingPolicyEventHandlerDefinitionTest |
| 4  | java↔java         | 416.250000       | 5         | CommandBusTestUtils↔AggregateTestFixture                                                      |
| 5  | java↔java         | 370.000000       | 6         | HandlerComparator↔AnnotationSagaMetaModelFactory                                              |
| 6  | java↔java         | 317.142857       | 7         | ExceptionHandlerTest↔AnnotationSagaMetaModelFactory                                           |
| 7  | java↔java         | 277.500000       | 8         | FieldChildEntityFieldDefinition↔GetterEvolverChildEntityFieldDefinition                       |
| 8  | java↔java         | 277.500000       | 9         | CommandBusTestUtils↔SimpleCommandBusTest                                                      |
| 9  | java↔java         | 266.400000       | 10        | SimpleEventHandlingComponentSequencingPolicyTest↔SimpleEventHandlingComponentTest             |
| 10 | xml↔xml           | 92.500000        | 1         | pom↔pom                                                                                       |
| 11 | xml↔xml           | 71.153846        | 2         | pom↔pom                                                                                       |
| 12 | xml↔xml           | 51.388889        | 3         | pom↔pom                                                                                       |
| 13 | xml↔xml           | 47.435897        | 4         | pom↔pom                                                                                       |
| 14 | xml↔xml           | 42.692308        | 5         | pom↔pom                                                                                       |
| 15 | xml↔xml           | 42.633745        | 6         | pom↔pom                                                                                       |
| 16 | xml↔xml           | 41.111111        | 7         | pom↔pom                                                                                       |
| 17 | xml↔xml           | 41.111111        | 8         | pom↔pom                                                                                       |
| 18 | xml↔xml           | 37.000000        | 9         | pom↔pom                                                                                       |
| 19 | xml↔xml           | 36.543210        | 10        | pom↔pom                                                                                       |
| 20 | java↔md           | 22.965517        | 1         | CommandConverter↔modularity                                                                   |
| 21 | java↔md           | 7.235984         | 2         | NoAppendConditionTest↔design-principles                                                       |
| 22 | java↔md           | 6.776557         | 3         | NoAppendCondition↔design-principles                                                           |
| 23 | java↔md           | 6.324786         | 4         | DefaultAppendConditionTest↔design-principles                                                  |
| 24 | java↔md           | 5.692308         | 5         | AppendCondition↔design-principles                                                             |
| 25 | java↔md           | 4.851870         | 6         | QueryThreadingIntegrationTest↔README                                                          |
| 26 | java↔md           | 4.833091         | 7         | AxonServerMessageStream↔design-principles                                                     |
| 27 | java↔md           | 4.065934         | 8         | AccessSerializingRepository↔design-principles                                                 |
| 28 | java↔md           | 3.940828         | 9         | DefaultSourcingConditionTest↔design-principles                                                |
| 29 | java↔md           | 3.881119         | 10        | SourcingConditionTest↔design-principles                                                       |
| 30 | java↔yaml         | 123.333333       | 1         | RenameCourseController↔application                                                            |
| 31 | java↔yaml         | 111.000000       | 2         | SubscribeStudentToCourseController↔application                                                |
| 32 | java↔yaml         | 74.000000        | 3         | UpdateCheckerProperties↔application                                                           |
| 33 | java↔yaml         | 65.294118        | 4         | UpdateCheckerAutoConfiguration↔application                                                    |
| 34 | java↔yaml         | 46.250000        | 5         | EventProcessingAutoConfiguration↔application                                                  |
| 35 | java↔yaml         | 39.642857        | 6         | EventProcessingAutoConfiguration↔test-docker-compose                                          |
| 36 | java↔yaml         | 11.482759        | 7         | SpringUtils↔slack-release-notification                                                        |
| 37 | java↔yaml         | 6.068890         | 8         | NoAppendConditionTest↔add-to-project                                                          |
| 38 | java↔yaml         | 5.967742         | 9         | AppendCondition↔add-to-project                                                                |
| 39 | java↔yaml         | 5.683564         | 10        | NoAppendCondition↔add-to-project                                                              |

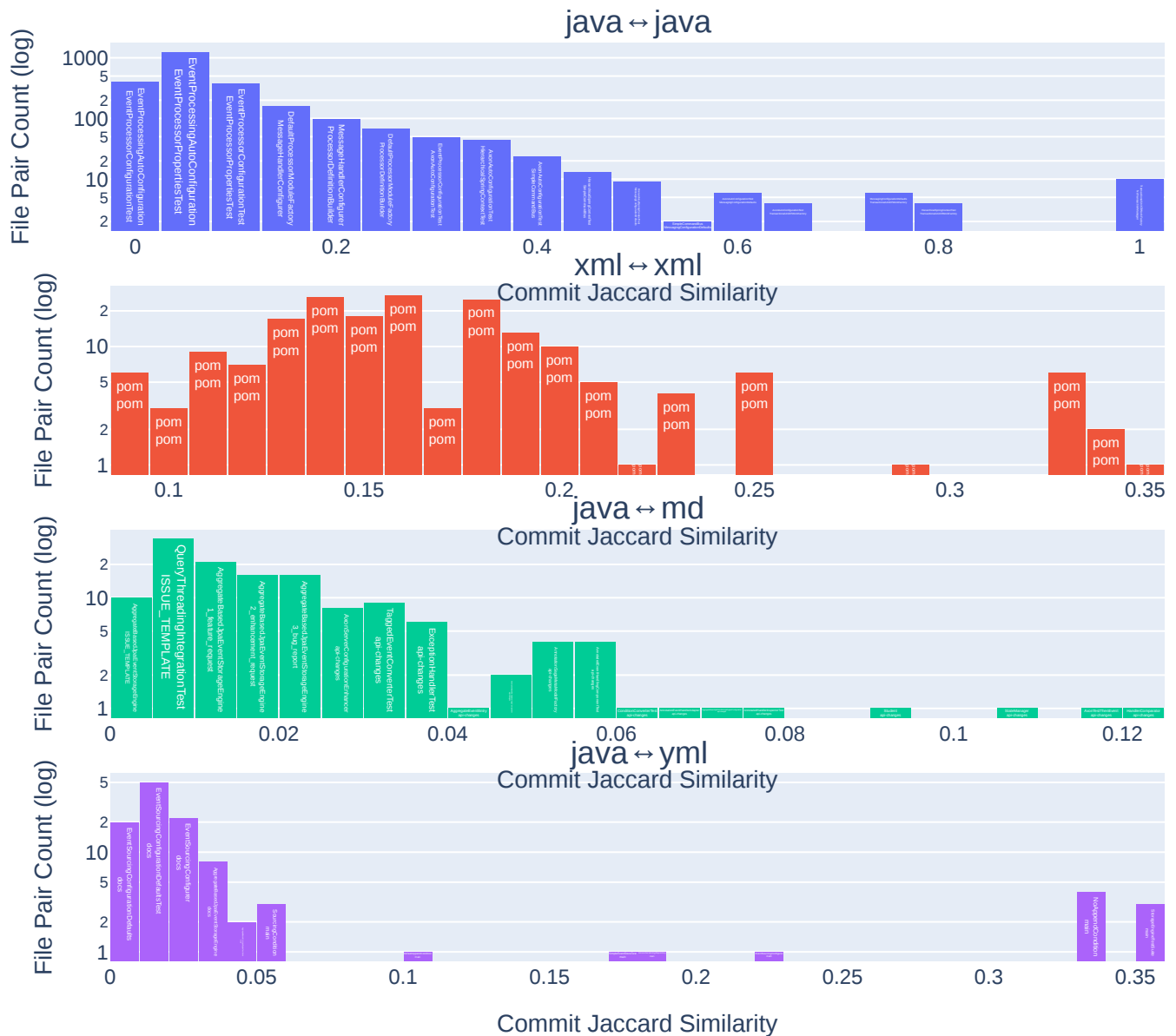
# Co-Changed Files by their commit lift



## Files changed together by commit Jaccard similarity

| fileExtensionPair | updateCommitJaccardSimilarity | GroupRank |                                                                                    |
|-------------------|-------------------------------|-----------|------------------------------------------------------------------------------------|
| 0                 | java↔java                     | 1.000000  | 1 Com                                                                              |
| 1                 | java↔java                     | 0.818182  | 2 ChildEntityFieldDefinition↔GetterSetterChildEntityFi                             |
| 2                 | java↔java                     | 0.800000  | 3 MethodSequencingPolicyEventHandlerDefinition↔MethodSequencingPolicyEventHandlerC |
| 3                 | java↔java                     | 0.750000  | 4 DefaultProcessorModuleFactory↔ProcessorDefi                                      |
| 4                 | java↔java                     | 0.666667  | 5 HandlerComparator↔AnnotationSagaMeta                                             |
| 5                 | java↔java                     | 0.636364  | 6 ChildEntityFieldDefinition↔FieldChildEntityFi                                    |
| 6                 | java↔java                     | 0.600000  | 7 CommandBusTestUtils↔Aggrega                                                      |
| 7                 | java↔java                     | 0.571429  | 8 ExceptionHandlerTest↔AnnotationSagaMeta                                          |
| 8                 | java↔java                     | 0.500000  | 9 MergeTa                                                                          |
| 9                 | java↔java                     | 0.488636  | 10 DefaultSourcingConditionTest↔SourcingC                                          |
| 10                | xml↔xml                       | 0.350000  | 1                                                                                  |
| 11                | xml↔xml                       | 0.341463  | 2                                                                                  |
| 12                | xml↔xml                       | 0.333333  | 3                                                                                  |
| 13                | xml↔xml                       | 0.285714  | 4                                                                                  |
| 14                | xml↔xml                       | 0.250000  | 5                                                                                  |
| 15                | xml↔xml                       | 0.227273  | 6                                                                                  |
| 16                | xml↔xml                       | 0.219512  | 7                                                                                  |
| 17                | xml↔xml                       | 0.214286  | 8                                                                                  |
| 18                | xml↔xml                       | 0.209302  | 9                                                                                  |
| 19                | xml↔xml                       | 0.200000  | 10                                                                                 |
| 20                | java↔md                       | 0.122682  | 1 AggregateBasedAxonServerEventStorageEngine↔                                      |
| 21                | java↔md                       | 0.116000  | 2 AggregateBasedStorageEngineTestSuite↔                                            |
| 22                | java↔md                       | 0.107143  | 3 AggregateBasedJpaEventStorageEngine↔                                             |
| 23                | java↔md                       | 0.091864  | 4 StorageEngineTestSuite↔                                                          |
| 24                | java↔md                       | 0.077247  | 5 SimpleEventStoreTest↔                                                            |
| 25                | java↔md                       | 0.071529  | 6 DefaultEventStoreTransactionTest↔                                                |
| 26                | java↔md                       | 0.066667  | 7 QueryThreadingIntegrationTest↔                                                   |
| 27                | java↔md                       | 0.062500  | 8 NoAppendConditionTest↔desi                                                       |
| 28                | java↔md                       | 0.059524  | 9 NoAppendCondition↔desi                                                           |
| 29                | java↔md                       | 0.057665  | 10 DefaultEventStoreTransaction↔                                                   |
| 30                | java↔yaml                     | 0.357143  | 1 RenameCourseController                                                           |
| 31                | java↔yaml                     | 0.333333  | 2 SubscribeStudentToCourseController                                               |
| 32                | java↔yaml                     | 0.227273  | 3 UpdateCheckerAutoConfiguration                                                   |
| 33                | java↔yaml                     | 0.187500  | 4 UpdateCheckerProperties                                                          |
| 34                | java↔yaml                     | 0.172414  | 5 EventProcessingAutoConfiguration                                                 |
| 35                | java↔yaml                     | 0.107143  | 6 EventProcessingAutoConfiguration↔test-doc                                        |
| 36                | java↔yaml                     | 0.058824  | 7 NoAppendConditionTest↔ai                                                         |
| 37                | java↔yaml                     | 0.058140  | 8 AppendCondition↔ai                                                               |
| 38                | java↔yaml                     | 0.056180  | 9 NoAppendCondition↔ai                                                             |
| 39                | java↔yaml                     | 0.049383  | 10 DefaultAppendConditionTest↔ai                                                   |

## Co-Changed Files by their commit jaccard similarity



## Find pairwise changed files with many highly ranked metrics

Find those pairwise changed files that have a high rank in many metrics by calculating a combined (weighted) score based on the ranks of each metric. This is useful to identify pairs of files that score high in most metrics, which indicates a strong co-change relationship.

|    | fileExtensionPair | filePair                                                                               | combinedMetricsScore | updateCommitCountEx |
|----|-------------------|----------------------------------------------------------------------------------------|----------------------|---------------------|
| 0  | java↔java         | Command↔Event                                                                          | 60                   |                     |
| 1  | java↔java         | Event↔Query                                                                            | 60                   |                     |
| 2  | java↔java         | Command↔Query                                                                          | 60                   |                     |
| 3  | java↔java         | Query↔QueryResponse                                                                    | 60                   |                     |
| 4  | java↔java         | Command↔QueryResponse                                                                  | 60                   |                     |
| 5  | java↔java         | Event↔QueryResponse                                                                    | 60                   |                     |
| 6  | java↔java         | FieldChildEntityFieldDefinitionTest↔GetterEvolverChildEntityFieldDefinitionTest        | 60                   |                     |
| 7  | java↔java         | FieldChildEntityFieldDefinitionTest↔GetterSetterChildEntityFieldDefinitionTest         | 60                   |                     |
| 8  | java↔java         | GetterEvolverChildEntityFieldDefinitionTest↔GetterSetterChildEntityFieldDefinitionTest | 60                   |                     |
| 9  | java↔java         | FieldChildEntityFieldDefinition↔GetterEvolverChildEntityFieldDefinition                | 62                   |                     |
| 10 | java↔md           | AggregateBasedAxonServerEventStorageEngine↔api-changes                                 | 28                   |                     |
| 11 | java↔md           | AggregateBasedJpaEventStorageEngine↔api-changes                                        | 39                   |                     |
| 12 | java↔md           | AggregateBasedStorageEngineTestSuite↔api-changes                                       | 40                   |                     |
| 13 | java↔md           | CommandConverter↔modularity                                                            | 53                   |                     |
| 14 | java↔md           | SimpleEventStoreTest↔api-changes                                                       | 57                   |                     |
| 15 | java↔md           | QueryThreadingIntegrationTest↔api-changes                                              | 59                   |                     |
| 16 | java↔md           | QueryThreadingIntegrationTest↔README                                                   | 63                   |                     |
| 17 | java↔md           | AxonServerEventStorageEngine↔api-changes                                               | 67                   |                     |
| 18 | java↔md           | DefaultEventStoreTransactionTest↔api-changes                                           | 70                   |                     |
| 19 | java↔md           | StorageEngineTestSuite↔api-changes                                                     | 71                   |                     |
| 20 | java↔yaml         | RenameCourseController↔application                                                     | 19                   |                     |
| 21 | java↔yaml         | ChangeCourseCapacityController↔application                                             | 19                   |                     |
| 22 | java↔yaml         | CreateCourseController↔application                                                     | 19                   |                     |
| 23 | java↔yaml         | SubscribeStudentToCourseController↔application                                         | 22                   |                     |
| 24 | java↔yaml         | EnrollStudentInFacultyController↔application                                           | 22                   |                     |
| 25 | java↔yaml         | CourseStatsController↔application                                                      | 22                   |                     |
| 26 | java↔yaml         | UnsubscribeStudentToCourseController↔application                                       | 22                   |                     |
| 27 | java↔yaml         | UpdateCheckerAutoConfiguration↔application                                             | 25                   |                     |
| 28 | java↔yaml         | EventProcessingAutoConfiguration↔application                                           | 28                   |                     |
| 29 | java↔yaml         | UpdateCheckerProperties↔application                                                    | 29                   |                     |
| 30 | xml↔xml           | pom↔pom                                                                                | 9                    |                     |
| 31 | xml↔xml           | pom↔pom                                                                                | 11                   |                     |
| 32 | xml↔xml           | pom↔pom                                                                                | 11                   |                     |
| 33 | xml↔xml           | pom↔pom                                                                                | 14                   |                     |
| 34 | xml↔xml           | pom↔pom                                                                                | 14                   |                     |
| 35 | xml↔xml           | pom↔pom                                                                                | 14                   |                     |
| 36 | xml↔xml           | pom↔pom                                                                                | 14                   |                     |
| 37 | xml↔xml           | pom↔pom                                                                                | 14                   |                     |
| 38 | xml↔xml           | pom↔pom                                                                                | 14                   |                     |
| 39 | xml↔xml           | pom↔pom                                                                                | 19                   |                     |

## Pairwise changed files with pareto-optimal metrics

A pair (count, confidence, jaccard, lift) is Pareto-optimal if there is no other pair that is better or equal in all metrics and strictly better in at least one. In other words, it is not "dominated" by any other pair.

The frontier = the “best tradeoffs.”

## Pairwise changed files with pareto-optimal metrics - not considering file extensions

|    | filePair                                                                               | combinedMetricsScore | updateCommitCount | updateCommitH |
|----|----------------------------------------------------------------------------------------|----------------------|-------------------|---------------|
| 0  | pom↔pom                                                                                | 9                    | 14                |               |
| 1  | AbstractConsistencyMarker↔ConsistencyMarker                                            | 343                  | 10                |               |
| 2  | ConsistencyMarker↔ConsistencyMarkers                                                   | 289                  | 17                |               |
| 3  | AggregateBasedAxonServerEventStorageEngine↔AggregateBasedJpaEventStorageEngine         | 668                  | 112               |               |
| 4  | EventSourcingConfigurationDefaults↔EventSourcingConfigurationDefaultsTest              | 731                  | 53                |               |
| 5  | StorageEngineTestSuite↔AggregateBasedStorageEngineTestSuite                            | 856                  | 114               |               |
| 6  | DefaultAppendConditionTest↔NoAppendConditionTest                                       | 222                  | 35                |               |
| 7  | DefaultSourcingConditionTest↔SourcingConditionTest                                     | 232                  | 43                |               |
| 8  | Command↔Event                                                                          | 60                   | 4                 |               |
| 9  | Event↔Query                                                                            | 60                   | 4                 |               |
| 10 | Command↔Query                                                                          | 60                   | 4                 |               |
| 11 | Query↔QueryResponse                                                                    | 60                   | 4                 |               |
| 12 | Command↔QueryResponse                                                                  | 60                   | 4                 |               |
| 13 | Event↔QueryResponse                                                                    | 60                   | 4                 |               |
| 14 | FieldChildEntityFieldDefinition↔GetterEvolverChildEntityFieldDefinition                | 62                   | 8                 |               |
| 15 | ChildEntityFieldDefinition↔GetterSetterChildEntityFieldDefinition                      | 73                   | 9                 |               |
| 16 | FieldChildEntityFieldDefinitionTest↔GetterEvolverChildEntityFieldDefinitionTest        | 60                   | 3                 |               |
| 17 | FieldChildEntityFieldDefinitionTest↔GetterSetterChildEntityFieldDefinitionTest         | 60                   | 3                 |               |
| 18 | GetterEvolverChildEntityFieldDefinitionTest↔GetterSetterChildEntityFieldDefinitionTest | 60                   | 3                 |               |

## Pairwise changed files with pareto-optimal metrics - using ranks grouped by file extensions

|   | fileExtensionPair | filePair                                                                   | combinedMetricsScore | updateCommitCount | updateCommitH |
|---|-------------------|----------------------------------------------------------------------------|----------------------|-------------------|---------------|
| 0 | java↔java         | AggregateBasedJpaEventStorageEngine↔EventSourcingConfigurationDefaultsTest | 3230                 | 4                 |               |
| 1 | java↔java         | AxonServerCommandBusConnector↔AggregateBasedStorageEngineTestSuite         | 3257                 | 3                 |               |
| 2 | java↔java         | EventSourcingConfigurationDefaults↔QueryThreadingIntegrationTest           | 3209                 | 4                 |               |
| 3 | java↔java         | QueryThreadingIntegrationTest↔AnnotationBasedEntityEvolvingComponentTest   | 3183                 | 3                 |               |

## WordCloud of git authors

|   | word                | frequency |
|---|---------------------|-----------|
| 0 | Steven van Beelen   | 6273      |
| 1 | Allard Buijze       | 2928      |
| 2 | smcvb               | 822       |
| 3 | Mitchell Herrijgers | 741       |
| 4 | Mateusz Nowak       | 632       |
| 5 | Milan Savic         | 449       |
| 6 | Gerard Klijs        | 246       |
| 7 | Marc Gathier        | 231       |
| 8 | Simon Zambrovski    | 226       |
| 9 | sara pellegrini     | 163       |

### Wordcloud of git authors

